

CHM 122

Introductory Organic Chemistry

2 units

Course outline

- Organic Chemistry: History, Development and Importance
- Nomenclature and chemistry of Organic Compounds
- Homologous series, Functional groups, Isolation and Purification of Organic compounds
- Saturated and unsaturated hydrocarbons.
- Qualitative and Quantitative Organic Chemistry
- Stereochemistry.
- Structures of simple sugars, starch, peptides and proteins.
- Synthetic polymers.

A Brief History of Organic Chemistry

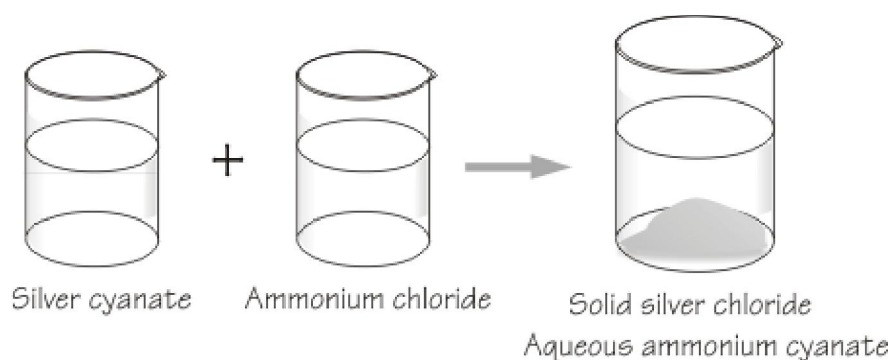
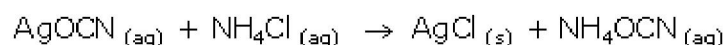
Organic chemistry is the study of compounds that contain carbon. It is one of the major branches of chemistry.

The history of organic chemistry can be traced back to ancient times when medicine men extracted chemicals from plants and animals to treat members of their tribes. They didn't label their work as "organic chemistry", they simply kept records of the useful properties and uses of things like willow bark which was used as a pain killer. (It is now known that willow bark contains acetylsalicylic acid, the ingredient in aspirin - chewing on the bark extracted the aspirin.) Their knowledge formed the basis of modern pharmacology which has a strong dependence on knowledge of organic chemistry.

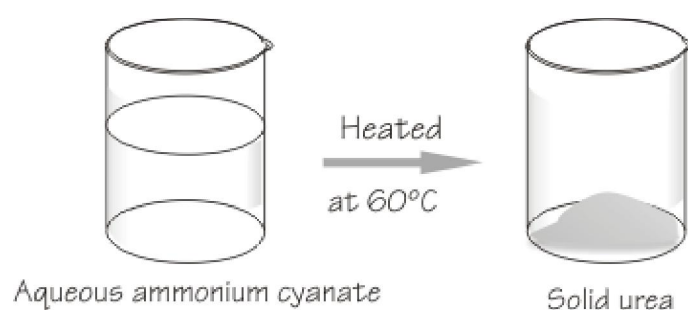
Organic chemistry was first defined as a branch of modern science in the early 1800's by Jon Jacob Berzelius. He classified chemical compounds into two main groups: organic if they originated in living or once-living matter and inorganic if they came from "mineral" or non-living matter. Like most chemists of his era, Berzelius believed in Vitalism - the idea that organic compounds could only originate from living organisms through the action of some vital force.

It was a student of Berzelius' who made the discovery that would result in the abandonment of Vitalism as a scientific theory. In 1828, Frederick Wöhler discovered that urea - an organic compound - could be made by heating ammonium cyanate (an inorganic compound).

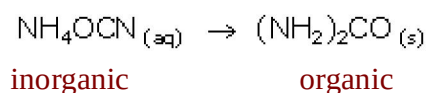
Wöhler mixed silver cyanate and ammonium chloride to produce solid silver chloride and aqueous ammonium cyanate:



He then separated the mixture by filtration and tried to purify the aqueous ammonium cyanate by evaporating the water.



To his surprise, the solid left over after the evaporation of the water was not ammonium cyanate, it was a substance with the properties of urea! Wöhler's observation marked the first time an organic compound had been synthesized from an inorganic source.



A Turning Point in Science History

Wöhler's discovery was a turning point in science history for two reasons. First, it undermined the idea of Vitalism because an organic compound was produced from an inorganic one. However, it also represented the discovery of isomerism - the possibility

of two or more different structures (ammonium cyanate crystals and urea crystals) based on the same chemical formula ($\text{N}_2\text{H}_4\text{CO}$).

Chemists started searching for reasons to explain isomerism. That in turn led to theories about the structure of chemical compounds. By the 1860's, chemists like Kékulé were proposing theories on the relationship between a compound's chemical formula and the physical distribution of its atoms. By the 1900's chemists were trying to determine the nature of chemical bonding by developing models for electron distribution. During all of this time the number of known organic compounds was increasing rapidly year by year.

During the 20th century, organic chemistry branched into sub-disciplines such as polymer chemistry, pharmacology, bioengineering, petro-chemistry, and numerous others. During that century, millions of new substances were discovered or synthesized. Today over 98% of all known compounds are organic.

Your study of organic chemistry begins at a time when the number of organic compounds and the number of reactions they undergo is nothing short of bewildering! Your study of organic chemistry begins with a study of the classification system, naming rules, and some key reactions that organic compounds undergo.

Sources of Organic Compounds

There are three generally accepted sources of organic compounds:

- carbonized organic matter
- living organisms
- invention/human ingenuity

Carbonized Organic Matter: Coal, Oil, and Natural Gas

Hundreds of millions of years ago, the organisms that inhabited earth were quite different than those we find here today. Plants were fast growing and lacked the woody tissues associated with the trees that currently dominate the world's productive ecosystems. Giant plants with broccoli-like stems grew rapidly, died, and decayed to form rich organic soils upon which more and more plants grew.

Eventually, thick layers of decomposing organic matter accumulated in much the same way that peat bogs do today. Over time these massive organic layers were buried under sediment, rock, or ice where they were subjected to tremendous pressures. In this way, they were transformed into various types of coal.

Meanwhile in Earth's prehistoric shallow seas, simple organisms like algae, bacteria and zooplankton thrived. As these tiny organisms died, they formed thick layers of organic matter on the sandy bottoms of these seas. Compression of layer upon layer of this material produced rocks known as shale. Under the tremendous pressures from the layers above, and with the shifting of earth's tectonic plates, the organic matter trapped in these rocks was converted to oil and natural gas over millions of years. The oil and gas migrated into porous rocks like sandstones or into large pockets of space located kilometres below the earth's surface. Thus organic matter from the past became today's fossil fuels.

Humans have known about fossil fuels for over 6000 years; however, only during the past 300 years have they been utilized on a large scale. Coal was the first of the fossil fuels to be extracted from the earth on a commercial basis. It was the fuel that drove the steam engines of the industrial revolution in the 18th, 19th, and 20th centuries.

Through a process called destructive distillation, coal was converted into coke, coal tar, and coal gas. Coke was used in the smelting of ores, coal tar was refined into over 200 different carbon compounds, and coal gas was used for things like street lighting!

Oil emerged as the dominant energy source for transportation in the 20th century. Natural gas is becoming the clean alternative to coal for generating electricity. It is also widely used as home heating and appliance fuel in North America. The economies of the western world are now completely dependent on oil and natural gas.

To some people, the burning of fossil fuels represents a tremendous waste. Not only does this practice contribute to the build up of carbon dioxide in the atmosphere, it also consumes that raw materials needed to make useful substances like plastics. By some estimates, the world will virtually exhaust its supply of oil and natural gas by 2050 - that's within your lifetime!

Nature: Living Organisms

Every living organism is a source of organic compounds. Each species is capable of producing a wide range of compounds, some of which are unique to that single species. The scent of a rose, the taste of a strawberry, and the fuzziness of a peach are the results of biochemical manufacturing processes within living things. Given that there are hundreds of thousands of species on earth, nature represents our most important source of organic compounds.

Humans have extracted and purified thousands of useful compounds from plants and animals. For example, the penicillin used to fight bacterial infections is extracted from a naturally occurring mold. Acetylsalicylic acid, commonly known as aspirin, comes from the bark of a species of willow tree. Vanilla flavouring is extracted from dried beans that come from a species of orchid called *Vanilla planifolia*. The heart drug digitalis comes from a plant called *Digitalis purpurea*. The list of examples goes on for volumes of pages.

Invention

Antibiotics, aspirin, vanilla flavouring, and heart drugs are examples of substances that no longer have to be obtained directly from nature. They are manufactured in laboratories from organic starting materials. Furthermore, experiments in which the chemical structures of naturally occurring substances are modified has produced organic compounds substances that do not exist anywhere in nature.

Each year over 250,000 new chemical compounds are discovered and many of these are products of scientists' imaginations, exploration, and in some cases - experiments gone wrong! Plastics are excellent examples of substances that are the product of invention - they are not found anywhere in nature.

Isolation and Purification of Organic Compounds

Organic compounds are ubiquitous hence they occur naturally in ponds, plants, animals, air and beneath the earth surface. They are usually found in impure form. The methods used in isolating organic compounds from its natural source of purifying it are determined mainly by the physical state of the compound. Sometimes, the chemical properties of a compound may be used to separate and purify it.

Organic gases or vapours are purified by methods involving the trapping of impurities (as for inorganic gases). For instance, ethane is purified by passing it through a potassium hydroxide (KOH) solution to remove acidic impurities such as SO_2 and CO_2 .

PURIFICATION: Before the properties and structure of an organic compound can be completely investigated, the compound must be pure. Common methods of purification are:-

- i. Recrystallization from suitable solvents.

- ii. Distribution: a) at atmospheric pressure; b) under reduced pressure or in vacuum; c) under increased pressure.
- iii. Steam distribution.
- iv. Sublimation.
- v. Chromatography; and
- vi. Zone melting.

There are various criteria for purity. The most common one for solids is m.p; for liquids b.p; density and refractive index are used; and more recently, the examination of the infra-red spectrum of a compound (whatever its normal physical state) has been used as a test for purity. In all cases, the process of purification is repeated until the physical constant or spectrum remains unchanged.

Before a natural product can be purified, it first has to be extracted from its source. Some of the earlier methods were solvent extraction, distillation and steam distillation. The methods used now are much superior, e.g. counter-current separation (this has replaced solvent extraction), electrophoresis, dialysis, diffusion, molecular distillation, ultracentrifugation, etc. the most important method nowadays is chromatography in its various forms. Zone melting is now also used for separating constituents in a mixture.

Organic solids are purified by simple crystallization. The aim of this method is to exterminate impurities and yield samples of the solid. At times a crystallized solid sample may have to be recrystallized (possibly three or four times) to obtain a pure sample of the solid. Evidence of the purity of a solid is provided by melting point determination, elemental analysis and spectra analysis (ultraviolet (UV), infra red (IR), nuclear magnetic resonance, NMR and mass spectrometry (MS)). A pure solid melts very sharply, at a definite temperature, passing from solid to liquid over a very small temperature range (say 0.1°C). On the other hand, an impure solid (a mixture) melts over a wide temperature range (probably several degrees).

For a sample containing two or more solids of interest together with contaminants, the method usually employed for separation and purification is fractional crystallization.

The purification of liquids is achieved mainly by simple distillation (for a single pure product) or fractional distillation (for separating two or more products from a complex mixture). Note that it is not always possible to purify a liquid completely by distillation alone. This is illustrated by ethanol – water mixture.

Other methods of purifying/separating organic compounds include steam distillation, solvent extraction and chromatography.

Steam distillation is commonly used for purifying aromatic compounds such as amino benzene (aniline), benzoic acid, be

The main Type of Organic Chemical

We think of a typical organic molecule as having carbon atoms joined together in a chain. The chain of carbon atoms makes up a backbone of the molecule. The hydrocarbon have a carbon backbone with only hydrogen atom joined to it, but in the other types of molecules somewhere along the chain there will be an atoms, or group of atoms, that give rise to the major properties of the molecule

Hydrocarbons that contain a delocalized ring of pi bonds such as the benzene ring are called aromatic hydrocarbons. Those that do not contain such delocalized systems are called aliphatic hydrocarbons.

Aliphatic hydrocarbons that contain only sigma (σ) bonds (i.e., only single bonds) are called saturated hydrocarbons. Those that contain both sigma and pi (π) bonds (i.e., double, triple, or delocalized bonds) are called unsaturated hydrocarbons.

A **functional group** is a special arrangement of atoms within an organic molecule that is responsible for some characteristic chemical behavior of the compound. Different molecules that contain the same functional groups have similar chemical behavior.

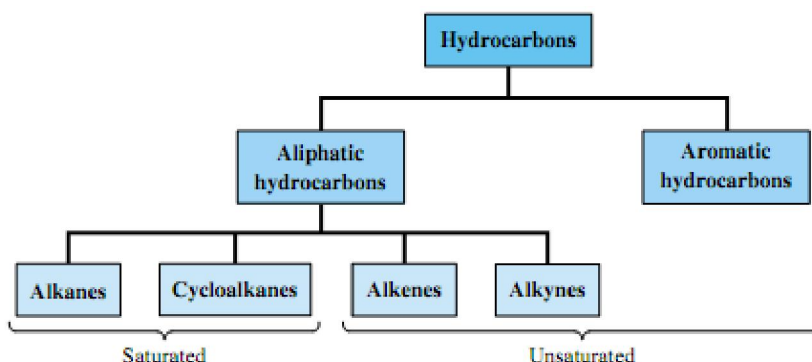


Figure showing Classification of hydrocarbons.

Saturated Hydrocarbons

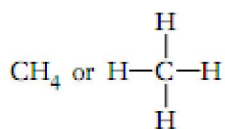
Alkanes and Cycloalkanes

The saturated hydrocarbons, or alkanes, are compounds in which each carbon atom is bonded to four other atoms. Cycloalkanes are saturated hydrocarbons that contain carbon atoms joined to form a ring.

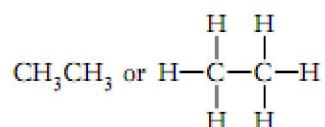
Alkanes

Alkanes have the general formula C_nH_{2n+2} where $n=1, 2, 3, \dots$. Each H atom in a saturated hydrocarbon is bonded to only one C atom. Saturated hydrocarbons contain only single bonds. There is no special functional group or reaction site on the molecule of an alkane. Sometimes, the alkanes are called paraffins because they are rather inert towards reagents such as acids, alkalis, oxidizing and reducing agents. Petroleum and natural gas are composed mostly of saturated hydrocarbons.

The IUPAC name of every member of the alkanes ends with the suffix-ane. The structure of the simplest alkane, methane, CH_4 . Methane molecules are tetrahedral with sp^3 hybridization at carbon.



Ethane, C_2H_6 , is the next simplest saturated hydrocarbon. Its structure is quite similar to that of methane. Two carbon atoms share a pair of electrons. Each carbon atom also shares an electron pair with each of three hydrogen atoms. Both carbon atoms are sp^3 hybridized.



Propane, C_3H_8 , is the next member of the family.

The first five members of the series are;

	CH_4	C_2H_6	C_3H_8	C_4H_{10}	C_5H_{12}
Number of C atoms = n	1	2	3	4	5
Number of H atoms = $2n + 2$	4	6	8	10	12

The formula of each alkane differs from the next by CH_2 , a methylene group.

A series of compounds in which each member differs from the next by a specific number and kind of atoms is called a **Homologous series**. It is a group of organic chemical compounds, usually listed in order of increasing size, that have a similar

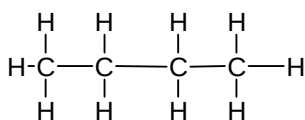
structure and whose structures differ only by the number of CH_2 units in the main carbon chain. The properties of members of such a series are closely related. As the molecular weights of the straight-chain hydrocarbons increase, their boiling points also increase regularly. Properties such as boiling point depend on the forces between molecules. Carbon-carbon and carbon-hydrogen bonds are essentially nonpolar and are arranged tetrahedrally around each C atom. As a result, saturated hydrocarbons are nonpolar molecules, and the only significant intermolecular forces are dispersion forces. These forces, which are due to induced dipoles, become stronger as the sizes of the molecules and the number of electrons in each molecule increase.

Sixteen Homologous series

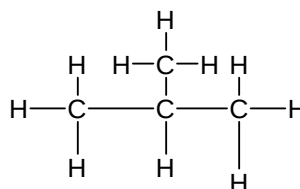
Series	Functional Group	General formula
Alkane	hydrocarbon chain	$\text{C}_n\text{H}_{2n+2}$
Alkene	Double bond	C_nH_{2n}
Alkyne	Triple bond	$\text{C}_n\text{H}_{2n-2}$
Halogen Alkane	Halogen atom	RX (X= Cl, Br etc)
Nitrile	cyanide group	RCN
Alcohol	OH group	ROH
Aldehyde	CHO	RCHO
Ketone	CO group	$\text{RR}'\text{CO}$
Acid	COOH group	RCOOH
Ester	COOC group	RCOOR'
Ether	COC group	ROR'
Amine	NH_2	RNH_2
Amide	CONH_2 group	RCONH_2
Acid chloride	COCl group	RCOCl
Arene	benzene ring	C_6H_6

Constitutional isomers (structured isomers) have the same molecular formula but differ in the order in which their atoms are attached.

For $n=4$, i.e. C_4H_{10} there are two isomers:



$H_3CCH_2CH_2CH_3$
butane



$CH_3CH(CH_3)CH_3$
2-methylpropane

For the C_5 hydrocarbons, there are three possible arrangements of the atoms. These three different C_5H_{12} alkanes are examples of constitutional isomers.

Some systematic method for naming compounds is necessary. The system in use today is prescribed by the International Union of Pure and Applied Chemistry (IUPAC). The names of the first 20 straight-chain alkanes are listed below. The names of the alkanes starting with pentane have prefixes (from Greek) that give the number of carbon atoms in the molecules. All alkane names have the -ane ending.

Molecular Formula	IUPAC Name	Normal bp (°C)	Normal mp (°C)	State at Room Temperature
CH_4	methane	-161	-184	gas
C_2H_6	ethane	-88	-183	
C_3H_8	propane	-42	-188	
C_4H_{10}	butane	+0.6	-138	
C_5H_{12}	pentane	36	-130	liquid
C_6H_{14}	hexane	69	-94	
C_7H_{16}	heptane	98	-91	
C_8H_{18}	octane	126	-57	
C_9H_{20}	nonane	150	-54	
$C_{10}H_{22}$	decane	174	-30	
$C_{11}H_{24}$	undecane	194.5	-25.6	
$C_{12}H_{26}$	dodecane	214.5	-9.6	
$C_{13}H_{28}$	tridecane	234	-6.2	
$C_{14}H_{30}$	tetradecane	252.5	+5.5	
$C_{15}H_{32}$	pentadecane	270.5	10	solid
$C_{16}H_{34}$	hexadecane	287.5	18	
$C_{17}H_{36}$	heptadecane	303	22.5	
$C_{18}H_{38}$	octadecane	317	28	
$C_{19}H_{40}$	nonadecane	330	32	
$C_{20}H_{42}$	eicosane	205 (at 15 torr)	36.7	

As the degree of branching increases for a series of molecules of the same molecular weight, the molecules become more compact. A compact molecule can have fewer points of contact with its neighbors than more extended molecules do. As a result, the total induced dipole forces (dispersion forces) are weaker for branched molecules, and the boiling points of such compounds are lower.

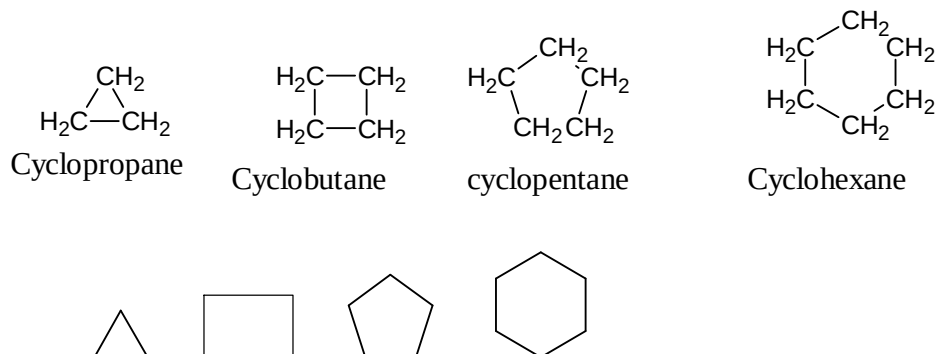
Isomeric C₆H₁₄ Alkanes

IUPAC Name	Formula	Normal bp (°C)
hexane	CH ₃ CH ₂ CH ₂ CH ₂ CH ₂ CH ₃	68.7
2-methylpentane	$\begin{array}{c} \text{CH}_3\text{CH}_2\text{CH}_2\text{CHCH}_3 \\ \\ \text{CH}_3 \end{array}$	60.3
3-methylpentane	$\begin{array}{c} \text{CH}_3\text{CH}_2\text{CHCH}_2\text{CH}_3 \\ \\ \text{CH}_3 \end{array}$	63.3
2,2-dimethylbutane	$\begin{array}{c} \text{CH}_3 \\ \\ \text{CH}_3\text{CH}_2\text{CCH}_3 \\ \\ \text{CH}_3 \end{array}$	49.7
2,3-dimethylbutane	$\begin{array}{c} \text{CH}_3\text{CH} - \text{CHCH}_3 \\ \quad \\ \text{CH}_3 \quad \text{CH}_3 \end{array}$	58.0
IUPAC Name	Formula	Normal bp (°C)
hexane	CH ₃ CH ₂ CH ₂ CH ₂ CH ₂ CH ₃	68.7
2-methylpentane	$\begin{array}{c} \text{CH}_3\text{CH}_2\text{CH}_2\text{CHCH}_3 \\ \\ \text{CH}_3 \end{array}$	60.3
3-methylpentane	$\begin{array}{c} \text{CH}_3\text{CH}_2\text{CHCH}_2\text{CH}_3 \\ \\ \text{CH}_3 \end{array}$	63.3
2,2-dimethylbutane	$\begin{array}{c} \text{CH}_3 \\ \\ \text{CH}_3\text{CH}_2\text{CCH}_3 \\ \\ \text{CH}_3 \end{array}$	49.7
2,3-dimethylbutane	$\begin{array}{c} \text{CH}_3\text{CH} - \text{CHCH}_3 \\ \quad \\ \text{CH}_3 \quad \text{CH}_3 \end{array}$	58.0

Cycloalkanes

The cyclic saturated hydrocarbons, or cycloalkanes, have the general formula C_nH_{2n}. The cycloalkanes (and other ring compounds that we will encounter later) are often shown in

simplified skeletal form in which each intersection of two lines represents a C atom; we mentally add enough H atoms to give each carbon atom four bonds. The first four unsubstituted cycloalkanes and their simplified representations are

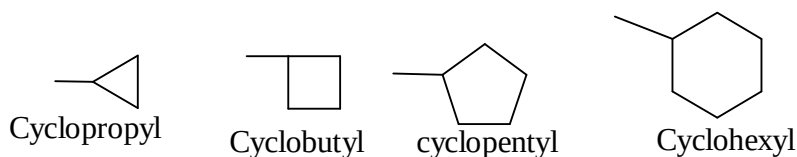


In some of these structures, the bond angles are somewhat distorted from the ideal tetrahedral angle of 109.5° , the most severe distortions being 60° in cyclopropane and 90° in cyclobutane. As a result, these rings are said to be “strained,” and these two compounds are unusually reactive for saturated hydrocarbons. Cyclopentane is stable with a nearly flat ring, because the bond angles in a regular pentagon (108°) are near the tetrahedral angle (109.5°). Cyclohexane is not stable as a flat ring. The bond angles of the flat ring would be 120° , not the 109.5° , of an sp^3 -hybridized carbon.

Naming Saturated Hydrocarbons

The IUPAC naming system is based on the names of the unbranched hydrocarbons and their higher homologues.

To name a branched-chain hydrocarbon, we first find the longest chain of carbon atoms and use the root name that corresponds to that hydrocarbon. We then indicate the positions and kinds of substituents attached to the chain. Alkyl group substituents attached to the longest chain are thought of as fragments of hydrocarbon molecules obtained by the removal of one hydrogen atom. They are given names related to the parent hydrocarbons from which they are derived. The general symbol R is to represent any alkyl group. The cycloalkyl groups derived from the first four cycloalkanes are called cyclopropyl, cyclobutyl, cyclopentyl, and cyclohexyl, respectively.



Some Alkanes and the Related Alkyl Groups

Parent Hydrocarbon	Alkyl group, -R	
CH_4 , methane $\begin{array}{c} \text{H} \\ \\ \text{H}-\text{C}-\text{H} \\ \\ \text{H} \end{array} \text{ or } \text{CH}_4$	$-\text{CH}_3$, methyl $\begin{array}{c} \text{H} \\ \\ -\text{C}-\text{H} \\ \\ \text{H} \end{array} \text{ or } -\text{CH}_3$	
C_2H_6 , ethane $\begin{array}{c} \text{H} \quad \text{H} \\ \quad \\ \text{H}-\text{C}-\text{C}-\text{H} \\ \quad \\ \text{H} \quad \text{H} \end{array} \text{ or } \text{CH}_3\text{CH}_3$	$-\text{C}_2\text{H}_5$, ethyl $\begin{array}{c} \text{H} \quad \text{H} \\ \quad \\ -\text{C}-\text{C}-\text{H} \\ \quad \\ \text{H} \quad \text{H} \end{array} \text{ or } -\text{CH}_2\text{CH}_3$	
C_3H_8 , propane $\begin{array}{c} \text{H} \quad \text{H} \quad \text{H} \\ \quad \quad \\ \text{H}-\text{C}-\text{C}-\text{C}-\text{H} \\ \quad \quad \\ \text{H} \quad \text{H} \quad \text{H} \end{array} \text{ or } \text{CH}_3\text{CH}_2\text{CH}_3$	$-\text{C}_3\text{H}_7$, propyl $\begin{array}{c} \text{H} \quad \text{H} \quad \text{H} \\ \quad \quad \\ -\text{C}-\text{C}-\text{C}-\text{H} \\ \quad \quad \\ \text{H} \quad \text{H} \quad \text{H} \end{array} \text{ or } -\text{CH}_2\text{CH}_2\text{CH}_3$	$-\text{C}_3\text{H}_7$, isopropyl $\begin{array}{c} \text{H} \quad \text{H} \quad \text{H} \\ \quad \quad \\ \text{H}-\text{C}-\text{C}-\text{C}-\text{H} \\ \quad \quad \\ \text{H} \quad \text{H} \quad \text{H} \end{array}$
C_4H_{10} , butane $\begin{array}{c} \text{H} \quad \text{H} \quad \text{H} \quad \text{H} \\ \quad \quad \quad \\ \text{H}-\text{C}-\text{C}-\text{C}-\text{C}-\text{H} \\ \quad \quad \quad \\ \text{H} \quad \text{H} \quad \text{H} \quad \text{H} \end{array} \text{ or } \text{CH}_3(\text{CH}_2)_2\text{CH}_3$	$-\text{C}_4\text{H}_9$, butyl $\begin{array}{c} \text{H} \quad \text{H} \quad \text{H} \quad \text{H} \\ \quad \quad \quad \\ -\text{C}-\text{C}-\text{C}-\text{C}-\text{H} \\ \quad \quad \quad \\ \text{H} \quad \text{H} \quad \text{H} \quad \text{H} \end{array} \text{ or } -\text{CH}_2(\text{CH}_2)_2\text{CH}_3$	$-\text{C}_4\text{H}_9$, <i>sec</i> -butyl (read as "secondary butyl") $\begin{array}{c} \text{H} \quad \text{H} \quad \text{H} \quad \text{H} \\ \quad \quad \quad \\ \text{H}-\text{C}-\text{C}-\text{C}-\text{C}-\text{H} \\ \quad \quad \quad \\ \text{H} \quad \text{H} \quad \text{H} \quad \text{H} \end{array}$
C_4H_{10} , methylpropane (common name, isobutane) $\begin{array}{c} \text{CH}_3 \\ \\ \text{H}-\text{C}-\text{CH}_3 \\ \\ \text{CH}_3 \end{array} \text{ or } \text{CH}(\text{CH}_3)_3$	$-\text{C}_4\text{H}_9$, <i>t</i> -butyl (read as "tertiary butyl") $\begin{array}{c} \text{CH}_3 \\ \\ -\text{C}-\text{CH}_3 \\ \\ \text{CH}_3 \end{array} \text{ or } -\text{C}(\text{CH}_3)_3$	$-\text{C}_4\text{H}_9$, isobutyl $\begin{array}{c} -\text{CH}_2 \\ \\ \text{H}-\text{C}-\text{CH}_3 \\ \\ \text{CH}_3 \end{array}$

Naming Organic Compounds

Functional group	Naming System
Alkene	suffix -ene
Alkyne	suffix -yne
Halogenalkane	prefix chloro- bromo- iodo-
Nitrile	suffix -nitrile

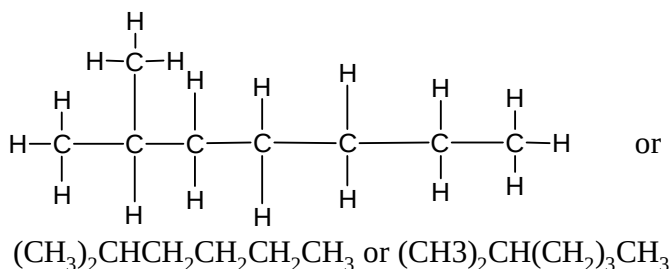
Alcohol	suffix	-ol
Aldehyde	suffix	-al
Ketone	suffix	-one
Acid	suffix	-oic acid
Ether	prefix	alkoxyl-
Amine	suffix	-amine
	Prefix	amino-
Amide	suffix	-amide
Acid chloride	suffix	oyl chloride

Nomenclature of Organic compound

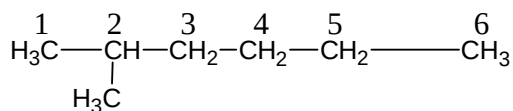
Summary of IUPAC Rules for Naming Alkanes

1. Find the longest chain of C atoms. Choose the base name that describes the number of C atoms in this chain, with the ending -ane. The longest chain may not be obvious at first if branches of different sizes are present.
2. Number the C atoms in this longest chain beginning at the end nearest the first branching. If necessary, go to the second branch closest to an end, and so on, until a difference is located. If there is branching at equal distances from both ends of the longest chain, begin numbering at the end nearest the branch that is first in alphabetical order.
3. Assign the name and position number to each substituent. Arrange the substituents in alphabetical order. Hyphenated prefixes, such as tert- and sec-, are not used in alphabetization of the substituents.
4. Use the appropriate prefix to group like substituents: di- 2, tri- 3, tetra- 4, penta- 5, and so on. Don't consider these prefixes when alphabetizing attached groups.
5. Write the name as a single word. Use hyphens to separate numbers and letters plus some hyphenated prefixes) and commas to separate numbers. Don't leave any spaces.

Example



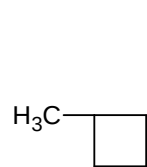
Following Rules 1 and 2 to number the carbon atoms in the longest chain.



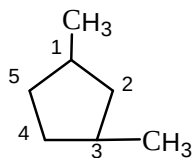
The methyl group is attached to the second carbon atom in a six-carbon chain, so the compound is named 2-methylhexane.

The names of substituted cycloalkanes are derived analogously to those of alkanes.

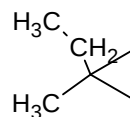
- (1) The base name is determined by the number of carbon atoms in the ring using the same base name as the alkane with the addition of cyclo in front.
- (2) If only one substituent is attached to the ring, no “location number” is required because all positions in a cycloalkane are equivalent.
- (3) Two or more functional groups on the ring are identified by location numbers, which should be assigned sequentially to the ring carbons in the order that gives the smallest sum of location numbers.



methylcyclobutane

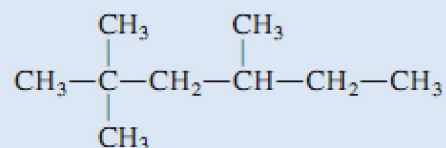


1,3-dimethylcyclopentane



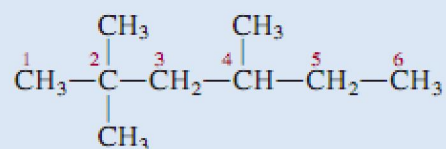
1-ethyl-1-methylcyclopropane

Give the IUPAC name of the following compound:



Strategy We follow the IUPAC rules and use the information in Table 11.2 to name the compound. How many C atoms are there in the longest chain?

Solution The longest chain has six C atoms so the parent compound is called hexane. Note that there are two methyl groups attached to carbon number 2 and one methyl group attached to carbon number 4.



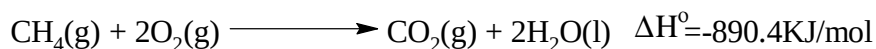
Therefore, we call the compound 2,2,4-trimethylhexane.

Properties of Alkanes

- The lowest member alkanes are colorless gases.
- Boiling Point: The boiling point of alkanes increases fairly regularly as the number of carbon atoms increase, but the rate of increase, decreases towards the higher members. The boiling point of isobutene is higher than that of n- butane. The higher the branching the lower the boiling point
- Melting Point: The melting point also increases but the variation is also irregular
- Solubility: Since alkane contain C –C single bond and C –H bonds all of which are non polar, we would expect alkane to have little or no interaction with hydrate polar solvent such as water. Hence, alkanes are insoluble in water e.g chloroform, methanol etc. However, as the solvent becomes less and less polar, solubility of alkanes increase e.g diethyl ether

Reactions of Alkanes

Combustion reactions: Alkanes are generally not considered to be very reactive substances. However, under suitable conditions they do react. For example, natural gas, gasoline, and fuel oil are alkanes that undergo highly exothermic

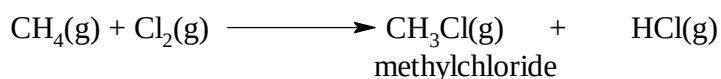


These, and similar combustion reactions, have long been utilized in industrial processes and in domestic heating and cooking.

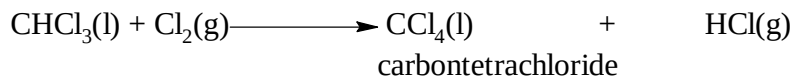
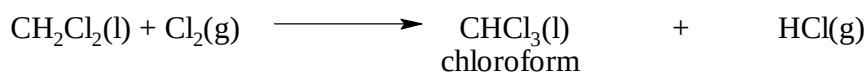
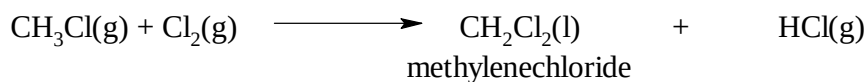
Catalytic reaction: Using $\text{Cr}_2\text{O}_7\cdot\text{Al}$ as catalyst at 700°C



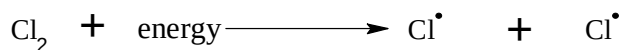
Halogenation of alkanes—that is, the replacement of one or more hydrogen atoms by halogen atoms—is another type of reaction that alkanes undergo. When a mixture of methane and chlorine is heated above 100°C or irradiated with light of a suitable wavelength, methyl chloride is produced:



If an excess of chlorine gas is present, the reaction can proceed further:

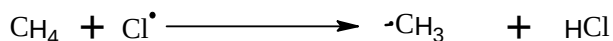


A great deal of experimental evidence suggests that the initial step of the first halogenation reaction occurs as follows:

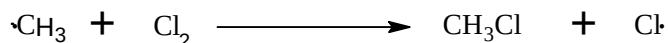


Thus, the covalent bond in Cl_2 breaks and two chlorine atoms form. We know it is the $\text{Cl}-\text{Cl}$ bond that breaks when the mixture is heated or irradiated because the bond enthalpy of Cl_2 is 242.7kJ/mol , whereas about 414kJ/mol are needed to break $\text{C}-\text{H}$ bonds in CH_4 .

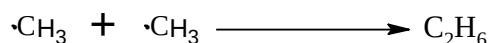
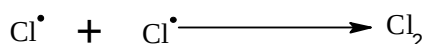
A chlorine atom is a radical, which contains an unpaired electron (shown by a single dot). Chlorine atoms are highly reactive and attack methane molecules according to the equation



This reaction produces hydrogen chloride and the methyl radical $\cdot\text{CH}_3$. The methyl radical is another reactive species; it combines with molecular chlorine to give methyl chloride and a chlorine atom:

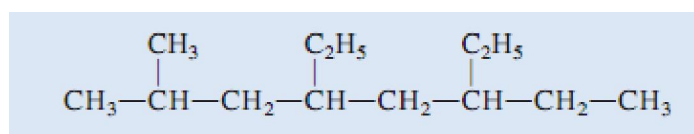


The production of methylene chloride from methyl chloride and any further reactions can be explained in the same way. The actual mechanism is more complex than the scheme shown because “side reactions” that do not lead to the desired products often take place, such as



Practice Exercise

Give the IUPAC name of the following compound



Write the structural formula of 3-ethyl-2,2-dimethylpentane.

Write the structural formula of 5-ethyl-2,6-dimethyloctane.

Unsaturated Hydrocarbons

There are three classes of unsaturated hydrocarbons: (1) the alkenes and their cyclic counterparts, the cycloalkenes; (2) the alkynes and the cycloalkynes; and (3) the aromatic hydrocarbons.

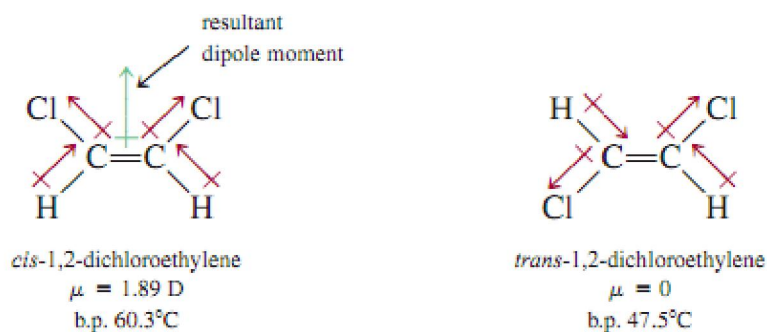
Alkenes

The simplest alkenes contain one carbon–carbon double bond, $C=C$, per molecule. The general formula for noncyclic alkenes is C_nH_{2n} . The simplest alkene is C_2H_4 , ethene, which is usually called by its common name, ethylene, in which both carbon atoms are sp^2 -hybridized and the double bond is made up of a sigma bond and a pi bond.

Geometric Isomers of Alkenes

In a compound such as ethane, C_2H_6 , the rotation of the two methyl groups about the carbon-carbon single bond (which is a sigma bond) is quite free. The situation is different for molecules that contain carbon-carbon double bonds, such as ethylene, C_2H_4 . In addition to the sigma bond, there is a pi bond between the two carbon atoms. Rotation about the carbon-carbon linkage does not affect the sigma bond, but it does move the two $2p_z$ orbitals out of alignment for overlap and, hence, partially or totally destroys the pi bond. This process requires an input of energy on the order of 270 kJ/mol. For this reason, the rotation of a carbon-carbon double bond is considerably restricted, but not impossible. Consequently, molecules containing carbon-carbon double bonds (that is, the alkenes) may have geometric isomers, which have the same type and number of atoms and the same chemical bonds but different spatial arrangements. Such isomers cannot be interconverted without breaking a chemical bond.

The molecule dichloroethylene, $ClHC=CHCl$, can exist as one of the two geometric isomers called *cis*-1,2-dichloroethylene and *trans*-1,2-dichloroethylene:



The existence of compounds with different arrangements of groups on the opposite sides of a bond with restricted rotation is called geometric isomerism. This *cis*–*trans* isomerism can occur across double bonds in alkenes and across single bonds in rings. Generally, *cis* and *trans* isomers have distinctly different physical and chemical properties. Heat or irradiation with light is commonly used to bring about the conversion of one geometric isomer to another, a process called *cis*–*trans* isomerization, or geometric isomerization.

Two shared electron pairs draw the atoms closer together than a single electron pair does. Thus, carbon–carbon double bonds are shorter than $C-C$ single bonds, $1.34 \text{ \AA}$ versus

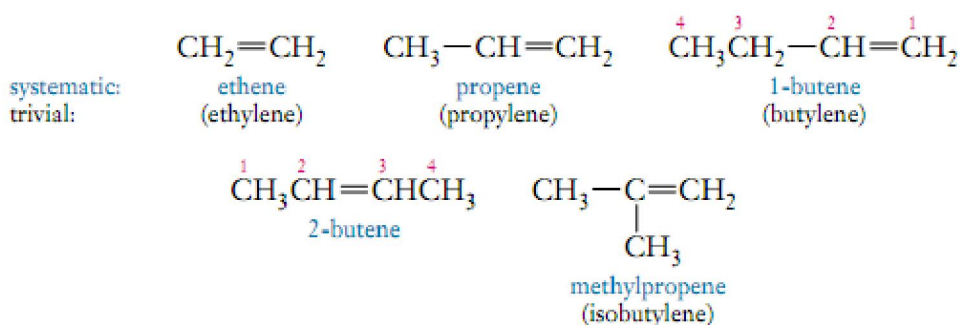
1.54 Å. The physical properties of the alkenes are similar to those of the alkanes, but their chemical properties are quite different.

The root for the name of each alkene is derived from the alkane having the same number of C atoms as the longest chain containing the double bond. In the trivial (common) system of nomenclature, the suffix -ylene is added to the characteristic root. In systematic (IUPAC) nomenclature, the suffix -ene is added to the characteristic root.

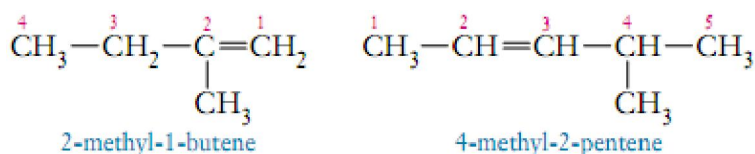
Summary of IUPAC Rules for Naming Alkenes and Cycloalkenes

1. Locate the C atoms in the longest C chain that contains the double bond. Use the base name prefix with the ending -ene.
2. Number the C atoms of this chain sequentially beginning at the end nearer the double bond. Insert the number describing the position of the double bond (indicated by its first carbon location) before the base name. (This is necessary only for chains of four or more C atoms, because only one position is possible for a double bond in a chain of two or three carbon atoms.)
3. In naming alkenes, the double bond takes positional precedence over substituents on the carbon chain. The double bond is assigned the lowest possible number.
4. To name compounds with possible geometric isomers, consider the two largest groups within the carbon chain that contains the double bond—these are indicated as part of the base name. The isomer in which the largest groups at each end of the C=C are located on opposite sides is called *trans*. If the largest groups are on the same side, the molecule is referred to as *cis*. Insert the prefix *cis*- or *trans*- just before the number of the double bond to indicate whether the largest groups are on the same or opposite sides, respectively, of the double bond.
5. For cycloalkenes, the double bond is assumed to be between C atoms 1 and 2, so no position number is needed to describe it.

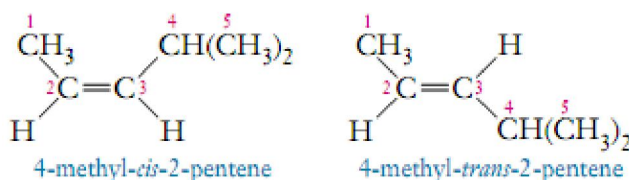
Some illustrations of this naming system follow.



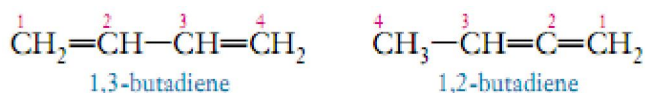
The following two names illustrate the application of Rule 3.



4-Methyl-2-pentene can exist as either of two isomers. Their names illustrate the application of Rule 4.



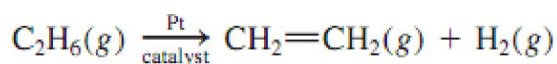
Some alkenes, called polyenes, have two or more carbon-carbon double bonds per molecule. The suffixes -adiene, -atriene, and so on are used to indicate the number of (C=C) double bonds in a molecule.



1,3-Butadiene and similar molecules that contain alternating single and double bonds are described as having conjugated double bonds. Such compounds are of special interest because of their polymerization reactions.

Properties and Reactions of Alkenes

Ethylene is an extremely important substance because it is used in large quantities in manufacturing organic polymers (very large molecules) and in preparing many other organic chemicals. Ethylene and other alkenes are prepared industrially by the cracking process, that is, the thermal decomposition of a large hydrocarbon into smaller molecules. When ethane is heated to about 800°C in the presence of platinum, it undergoes the following reaction:

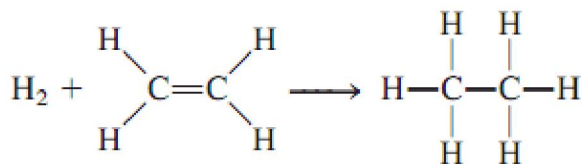


The platinum acts as a catalyst, a substance that speeds up a reaction without being used up in the process and therefore does not appear on either side of the equation.

Other alkenes can be prepared by cracking the higher members of the alkane family. Alkenes are classified as unsaturated hydrocarbons, compounds with double or triple carbon-carbon bonds.

Unsaturated hydrocarbons commonly undergo addition reactions in which one molecule adds to another to form a single product. An example of an addition reaction is hydrogenation, which is the addition of molecular hydrogen to compounds containing

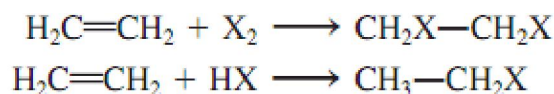
C=C and $\text{C}\equiv\text{C}$ bonds



Hydrogenation is an important process in the food industry. Vegetable oils have considerable nutritional value, but many oils must be hydrogenated to eliminate some of the C=C bonds before they can be used to prepare food. Upon exposure to air, polyunsaturated molecules—molecules with many C=C bonds—undergo oxidation to yield unpleasant-tasting products (vegetable oil that has oxidized is said to be rancid).

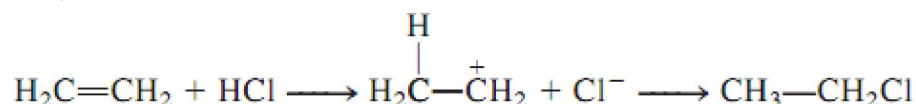
In the hydrogenation process, a small amount of nickel catalyst is added to the oil and the mixture is exposed to hydrogen gas at high temperature and pressure. Afterward, the nickel is removed by filtration. Hydrogenation reduces the number of double bonds in the molecule but does not completely eliminate them. If all the double bonds are eliminated, the oil becomes hard and brittle. Under controlled conditions, suitable cooking oils and margarine may be prepared by hydrogenation from vegetable oils extracted from cottonseed, corn, and soybeans.

Other addition reactions to the C=C bond involve the hydrogen halides and halogens

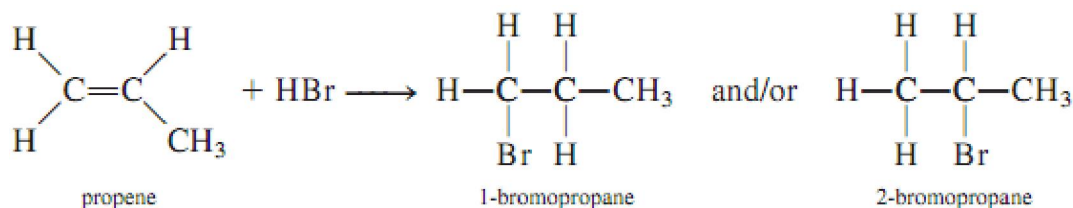


in which X represents a halogen atom.

For example; HCl and ethylene. When the two molecules react, the interaction is between the electron-rich region (pi electrons of the double bond) and the electron-poor region of HCl, which is the H atom. The steps are



The addition of a hydrogen halide to an unsymmetrical alkene such as propene is more complicated because two products are possible:



In reality, however, only 2-bromopropane is formed. This phenomenon was observed in all reactions between unsymmetrical reagents and alkenes. In 1871 the Russian chemist Vladimir Markovnikov postulated a generalization that enables us to predict the outcome of such an addition reaction. This generalization, now known as **Markovnikov's rule**, states that in the addition of unsymmetrical (that is, polar) reagents to alkenes, the positive portion of the reagent (usually hydrogen) adds to the carbon atom in the double bond that already has the most hydrogen atoms.

The C atom to which the two H atoms are attached has a higher electron density. Therefore, this is the site for the H^+ ion (from HBr) to form a C—H bond, followed by the formation of the C—Br bond on the other C atom. Finally we note that ethylene undergoes a different type of addition reaction that leads to the formation of a polymer. In this process, first an initiator molecule (R_2) is heated to produce two radicals:



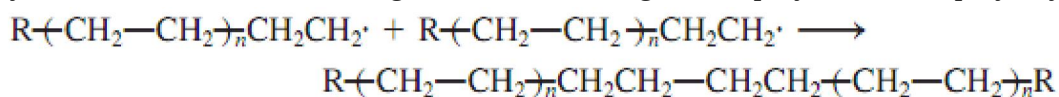
The reactive radical attacks an ethylene molecule to generate a new radical (it is the pi bond that is broken in the polymerization of ethylene):



which further reacts with another ethylene molecule, and so on:



Very quickly a long chain of CH_2 groups is built. Eventually, this process is terminated by the combination of two long-chain radicals to give the polymer called polyethylene

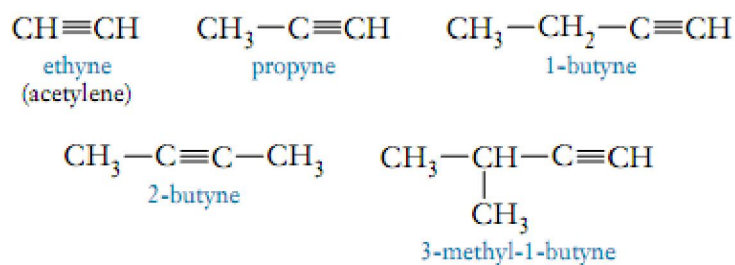


where $(-\text{CH}_2-\text{CH}_2-)$ is a convenient shorthand convention for representing the repeating unit in the polymer. The value of n is understood to be very large, on the order of thousands. Under different conditions, it is possible to prepare polyethylene with branched chains. Today, many different forms of polyethylene with widely different physical properties are known. Polyethylene is mainly used in films in frozen food packaging and other product wrappings.

Alkynes

The alkynes, or acetylenic hydrocarbons, contain one or more carbon–carbon triple bonds, $\text{—C}\equiv\text{C—}$. The noncyclic alkynes with one triple bond per molecule have the general formula $\text{C}_n\text{H}_{2n-2}$. The bonding in all alkynes is similar to that in acetylene. Triply bonded carbon atoms are sp -hybridized. The triply bonded atoms and their adjacent atoms lie on a straight line.

Alkynes are named like the alkenes except that the suffix -yne is added to the characteristic root. The first member of the series is commonly called acetylene. Its molecular formula is C_2H_2 . It is thermodynamically unstable, decomposing explosively to C(s) and $\text{H}_2\text{(g)}$ at high pressures. It may be converted into ethene (also known as ethylene) and then to ethane by the addition of hydrogen. The condensed formulas for several small alkynes are



The triple bond takes positional precedence over substituents on the carbon chain. It is assigned the lowest possible number in naming.

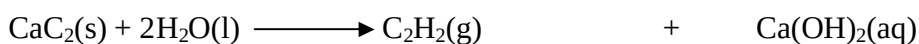
Summary of IUPAC Rules for Naming Alkynes

Alkynes are named like the alkenes except for the following two points.

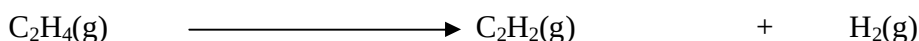
1. The suffix -yne is added to the characteristic root.
2. Because the linear arrangement about the triple bond does not lead to geometric isomerism, the prefixes cis- and trans- are not used.

Properties and Reactions of Alkynes

The simplest alkyne is ethyne, better known as acetylene (C_2H_2). Acetylene is a colorless gas (b.p. 284°C) prepared in the laboratory by the reaction between calcium carbide and water:



Industrially, it is prepared by the thermal decomposition of ethylene at about 1100°C:



Acetylene has many important uses in industry. Because of its high heat of combustion



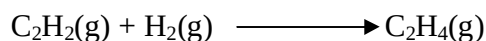
acetylene burned in an “oxyacetylene torch” gives an extremely hot flame (about 3000°C). Thus, oxyacetylene torches are used to weld metals.

Acetylene is unstable and has a tendency to decompose:

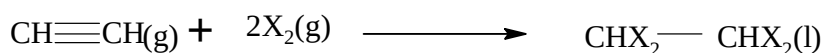
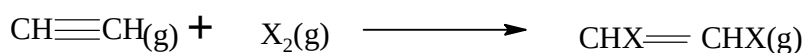
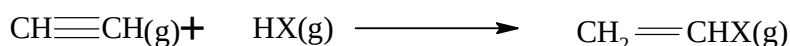


In the presence of a suitable catalyst or when the gas is kept under pressure, this reaction can occur with explosive violence. To be transported safely, it must be dissolved in an inert organic solvent such as acetone at moderate pressure. In the liquid state, acetylene is very sensitive to shock and is highly explosive.

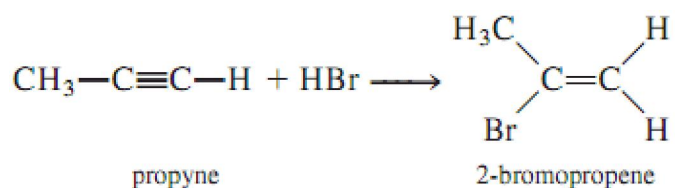
Being an unsaturated hydrocarbon, acetylene can be hydrogenated to yield ethylene:



It undergoes these addition reactions with hydrogen halides and halogens:



Methylacetylene (propyne), $\text{CH}_3-\text{C}\equiv\text{C}-\text{H}$, is the next member in the alkyne family. It undergoes reactions similar to those of acetylene. The addition reactions of propyne also obey Markovnikov's rule:



Aromatic Hydrocarbons

Originally the word “aromatic” was applied to pleasant-smelling substances. The word now describes benzene, its derivatives, and certain other compounds that exhibit similar chemical properties. Some have very foul odors because of substituents on the benzene ring. On the other hand, many fragrant compounds do not contain benzene rings.

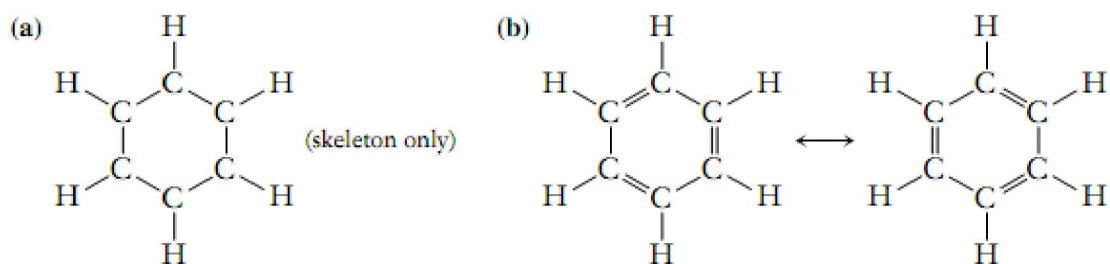
Steel production at one time required large amounts of coke. This was prepared by heating bituminous coal to high temperatures in the absence of air. This process also favors production of coal gas and coal tar. Because of the enormous amount of coal converted to coke, coal tar was produced in large quantities. It served as a source of aromatic compounds. For each ton of coal converted to coke, about 5 kg of aromatic compounds were obtained. Today petroleum refining is the major source of aromatic hydrocarbons. Early research on the reactions of the aromatic hydrocarbons led to methods for preparing a great variety of dyes, drugs, flavors, perfumes, and explosives. More recently, large numbers of polymeric materials, such as plastics and fabrics, have been prepared from these compounds.

Benzene

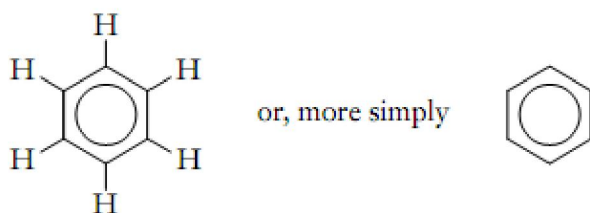
Benzene is the simplest aromatic hydrocarbon. By studying its reactions, we can learn a great deal about aromatic hydrocarbons. Benzene was discovered in 1825 by Michael Faraday when he fractionally distilled a by-product oil obtained in the manufacture of illuminating gas from whale oil.

Elemental analysis and determination of its molecular weight showed that the molecular formula for benzene is C_6H_6 . The formula suggests that it is highly unsaturated. But its properties are quite different from those of alkenes and alkynes.

The facts that only one monosubstitution product is obtained in many reactions and that no addition products can be prepared show conclusively that benzene has a symmetrical ring structure. Stated differently, every H atom is equivalent to every other H atom, and this is possible only in a symmetrical ring structure (a).

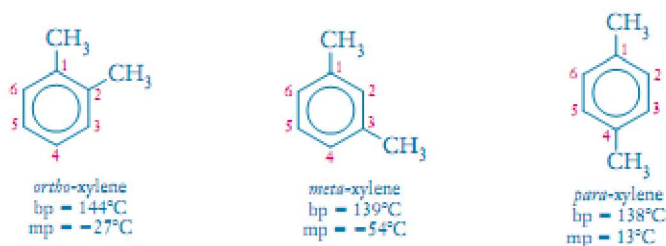


Friedrich Kekulé (1829–1896) suggested that the structure of benzene was intermediate between two structures [part (b) of the preceding structures] that we now call resonance structures. We often represent benzene as



Other Aromatic Hydrocarbons

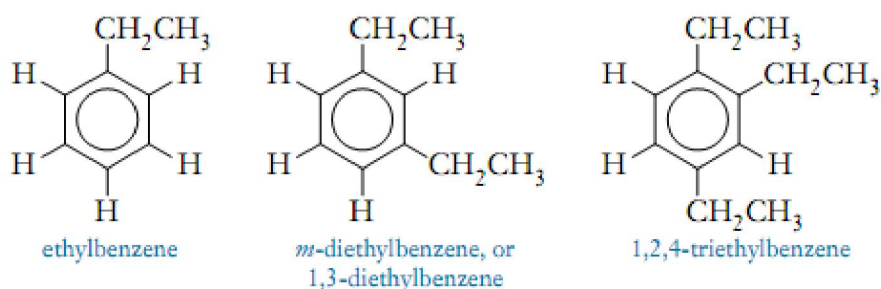
Benzene molecules bearing alkyl substituents are called alkylbenzenes. The simplest of these is methylbenzene (common name, toluene). The dimethylbenzenes are called xylenes. Three different compounds have the formula $C_6H_4(CH_3)_2$. These three xylenes are constitutional isomers. In naming these (as well as other disubstituted benzenes), we use prefixes ortho- (abbreviated o-), meta- (m-), or para- (p-) to refer to relative positions of substituents on the benzene ring. The ortho- prefix refers to two substituents located on adjacent carbon atoms; for example, 1,2-dimethylbenzene is o-xylene. The meta- prefix identifies substituents on C atoms 1 and 3, so 1,3-dimethylbenzene is m-xylene. The para- prefix refers to substituents on C atoms 1 and 4, so 1,4-dimethylbenzene is p-xylene.



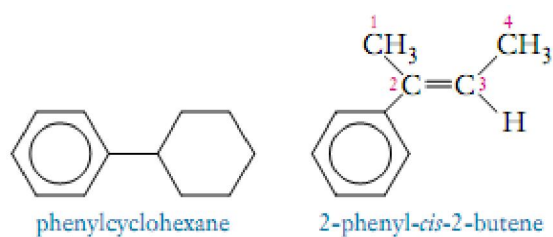
Summary of Rules for Naming Derivatives of Benzene

1. If there is only one group on the ring, no number is needed to designate its position.
2. If there are two groups on the ring, we use the traditional designations. ortho- or o- for 1,2-disubstitution meta- or m- for 1,3-disubstitution para- or p- for 1,4-disubstitution
3. If there are three or more groups on the ring, location numbers are assigned to give the minimum sum of numbers.

Examples are



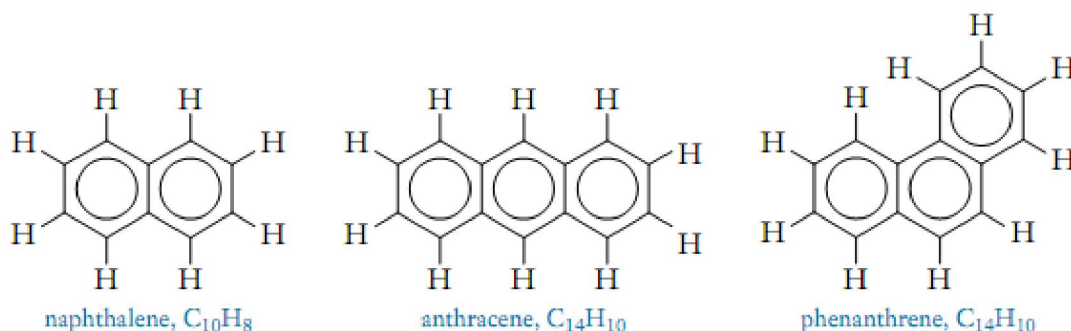
When an H atom is removed from a benzene molecule, C_6H_6 , the resulting group, C_6H_5 —or, is called “phenyl.” Sometimes we name mixed alkyl-aromatic hydrocarbons on that basis.



Aromatic Hydrocarbons from Coal Tar

Name	Formula	Normal bp (°C)	Normal mp (°C)	Solubility
benzene	C_6H_6	80	+6	
toluene	$C_6H_5CH_3$	111	−95	
<i>o</i> -xylene	$C_6H_4(CH_3)_2$	144	−27	
<i>m</i> -xylene	$C_6H_4(CH_3)_2$	139	−54	All insoluble in water
<i>p</i> -xylene	$C_6H_4(CH_3)_2$	138	+13	
naphthalene	$C_{10}H_8$	218	+80	
anthracene	$C_{14}H_{10}$	342	+218	
phenanthrene	$C_{14}H_{10}$	340	+101	

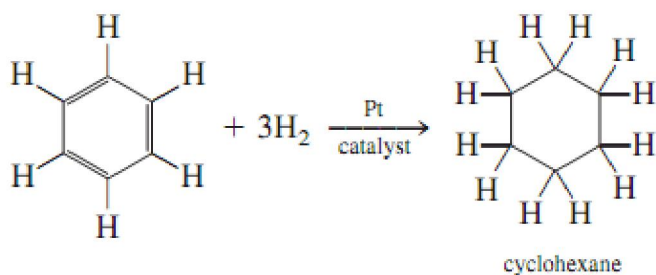
Another class of aromatic hydrocarbons consists of “condensed” or “fused-ring” aromatic systems. The simplest of these are naphthalene, anthracene, and phenanthrene.



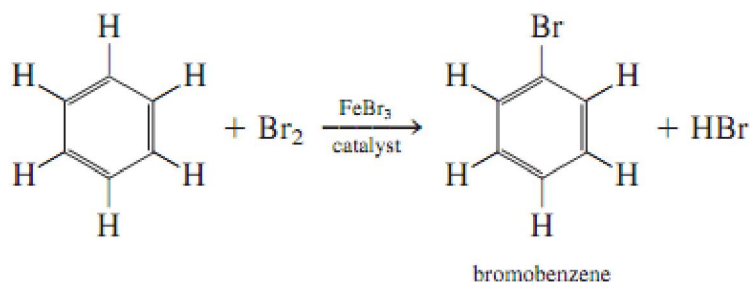
No hydrogen atoms are attached to the carbon atoms that are involved in fusion of aromatic rings, that is, carbon atoms that are members of two or more aromatic rings.

Properties and Reactions of Aromatic Compounds

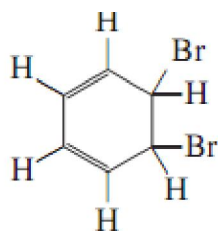
Benzene is a colorless, flammable liquid obtained chiefly from petroleum and coaltar. Perhaps the most remarkable chemical property of benzene is its relative inertness. Although it has the same empirical formula as acetylene (CH) and a high degree of unsaturation, it is much less reactive than either ethylene or acetylene. The stability of benzene is the result of electron delocalization. In fact, benzene can be hydrogenated, but only with difficulty. The following reaction is carried out at significantly higher temperatures and pressures than are similar reactions for the alkenes:



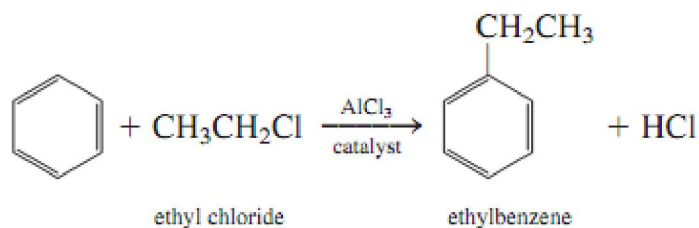
We saw earlier that alkenes react readily with halogens and hydrogen halides to form addition products, because the pi bond in C=C can be broken more easily. The most common reaction of halogens with benzene is substitution. For example,



Note that if the reaction were addition, electron delocalization would be destroyed in the product

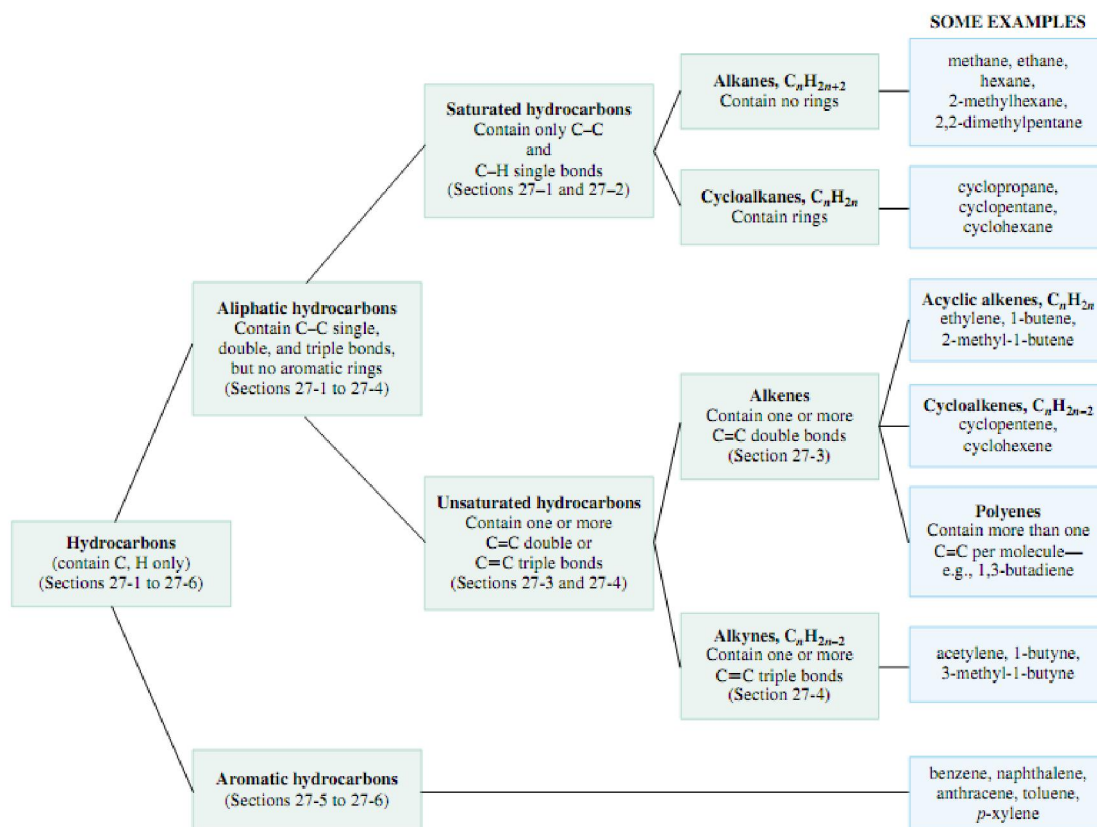


and the molecule would not have the aromatic characteristic of chemical unreactivity. Alkyl groups can be introduced into the ring system by allowing benzene to react with an alkyl halide using AlCl_3 as the catalyst:



An enormously large number of compounds can be generated from substances in which benzene rings are fused together. The best known of these compounds is naphthalene, which is used in mothballs. These and many other similar compounds are present in coal tar. Some of the compounds with several rings are powerful carcinogens—they can cause cancer in humans and other animals.

Summary of the classification of hydrocarbons

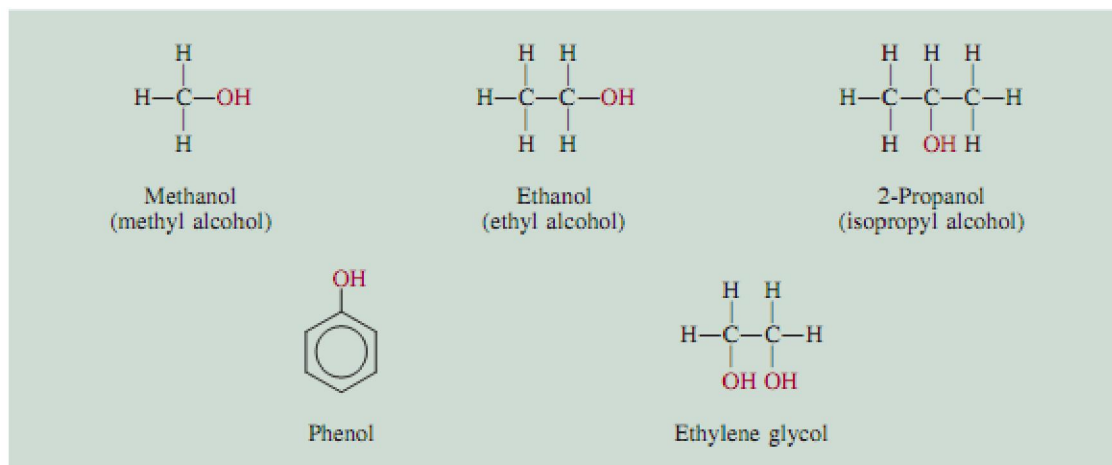
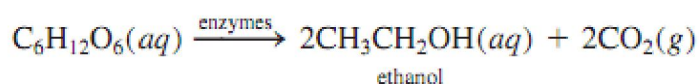


Chemistry of the Functional Groups

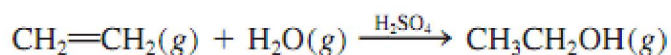
We now examine some organic functional groups, groups that are responsible for most of the reactions of the parent compounds. In particular, we focus on oxygen-containing and nitrogen-containing compounds.

Alcohols

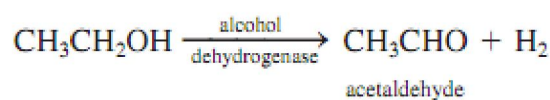
All alcohols contain the hydroxyl functional group, -OH. Some common alcohols are shown in Figure 11.17. Ethyl alcohol, or ethanol, is by far the best known. It is produced biologically by the fermentation of sugar or starch. In the absence of oxygen the enzymes present in bacterial cultures or yeast catalyze the reaction



Commercially, ethanol is prepared by an addition reaction in which water is combined with ethylene at about 280°C and 300 atm:

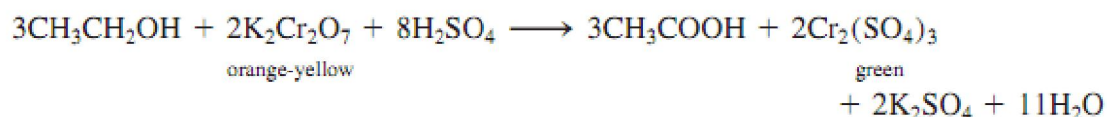


Ethanol has countless applications as a solvent for organic chemicals and as a starting compound for the manufacture of dyes, synthetic drugs, cosmetics, and explosives. It is also a constituent of alcoholic beverages. Ethanol is the only nontoxic (more properly, the least toxic) of the straight-chain alcohols; our bodies produce an enzyme, called alcohol dehydrogenase, which helps metabolize ethanol by oxidizing it to acetaldehyde:



This equation is a simplified version of what actually takes place; the H atoms are taken up by other molecules, so that no H₂ gas is evolved.

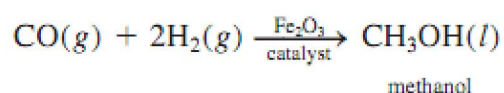
Ethanol can also be oxidized by inorganic oxidizing agents, such as acidified potassium dichromate, to acetic acid:



This reaction has been employed by law enforcement agencies to test drivers suspected of being drunk. A sample of the driver's breath is drawn into a device called a breath analyzer, where it is reacted with an acidic potassium dichromate solution.

From the color change (orange-yellow to green) it is possible to determine the alcohol content in the driver's blood.

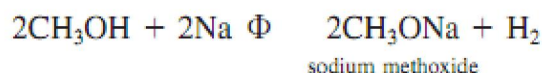
Ethanol is called an aliphatic alcohol because it is derived from an alkane (ethane). The simplest aliphatic alcohol is methanol, CH_3OH . Called wood alcohol, it was prepared at one time by the dry distillation of wood. It is now synthesized industrially by the reaction of carbon monoxide and molecular hydrogen at high temperatures and pressures:



Methanol is highly toxic. Ingestion of only a few milliliters can cause nausea and blindness. Ethanol intended for industrial use is often mixed with methanol to prevent people from drinking it. Ethanol containing methanol or other toxic substances is called denatured alcohol.

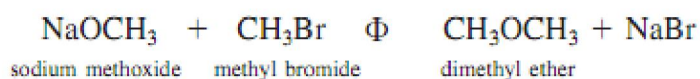
The alcohols are very weakly acidic; they do not react with strong bases, such as NaOH

. The alkali metals react with alcohols to produce hydrogen:

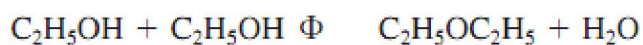


Ethers

Ethers contain the ROOR' linkage, where R and R' are a hydrocarbon (aliphatic or aromatic) group. They are formed by the reaction between an alkoxide (containing the RO^- ion) and an alkyl halide:

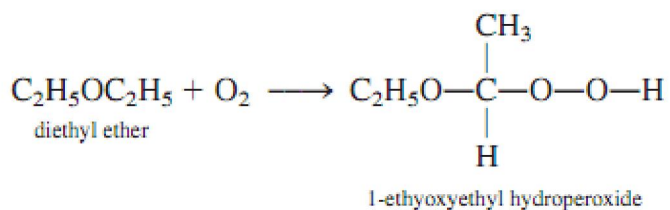


Diethyl ether is prepared on an industrial scale by heating ethanol with sulfuric acid at 140°C



This reaction is an example of a condensation reaction, which is characterized by the joining of two molecules and the elimination of a small molecule, usually water.

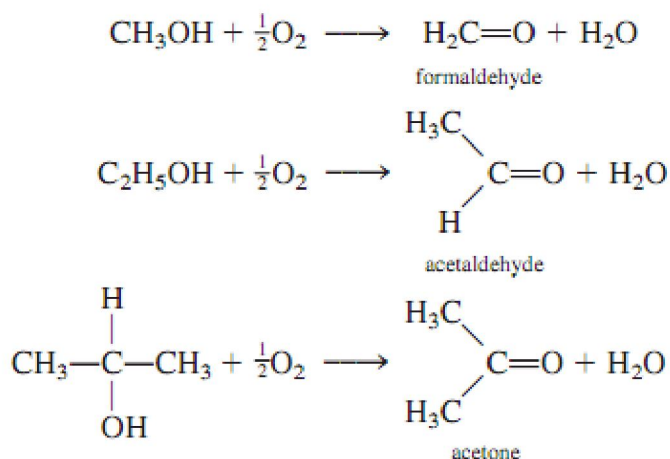
Like alcohols, ethers are extremely flammable. When left standing in air, they have a tendency to slowly form explosive peroxides:



Peroxides contain the —O—O— linkage; the simplest peroxide is hydrogen peroxide, H_2O_2 . Diethyl ether, commonly known as “ether,” was used as an anesthetic for many years. It produces unconsciousness by depressing the activity of the central nervous system. The major disadvantages of diethyl ether are its irritating effects on the respiratory system and the occurrence of postanesthetic nausea and vomiting. “Neothyl,” or methyl propyl ether, $\text{CH}_3\text{OCH}_2\text{CH}_2\text{CH}_3$, is currently favored as an anesthetic because it is relatively free of side effects.

Aldehydes and Ketones

Under mild oxidation conditions, it is possible to convert alcohols to aldehydes and ketones:

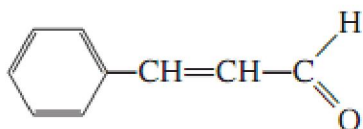


The functional group in these compounds is the carbonyl group,

>C=O . In an aldehyde at least one hydrogen atom is bonded to the carbon in the carbonyl group. In a ketone, the carbon atom in the carbonyl group is bonded to two hydrocarbon groups.

The simplest aldehyde, formaldehyde ($\text{H}_2\text{C}=\text{O}$) has a tendency to polymerize, i.e. the individual molecules join together to form a compound of high molar mass.

This action gives off much heat and is often explosive, so formaldehyde is usually prepared and stored in aqueous solution (to reduce the concentration). This rather disagreeable-smelling liquid is used as a starting material in the polymer industry and in the laboratory as a preservative for animal specimens. Interestingly, the higher molar mass aldehydes, such as cinnamic aldehyde



have a pleasant odor and are used in the manufacture of perfumes.

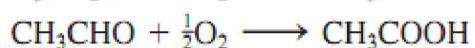
Ketones generally are less reactive than aldehydes. The simplest ketone is acetone, a pleasant-smelling liquid that is used mainly as a solvent for organic compounds

and nail polish remover.

Carboxylic Acids

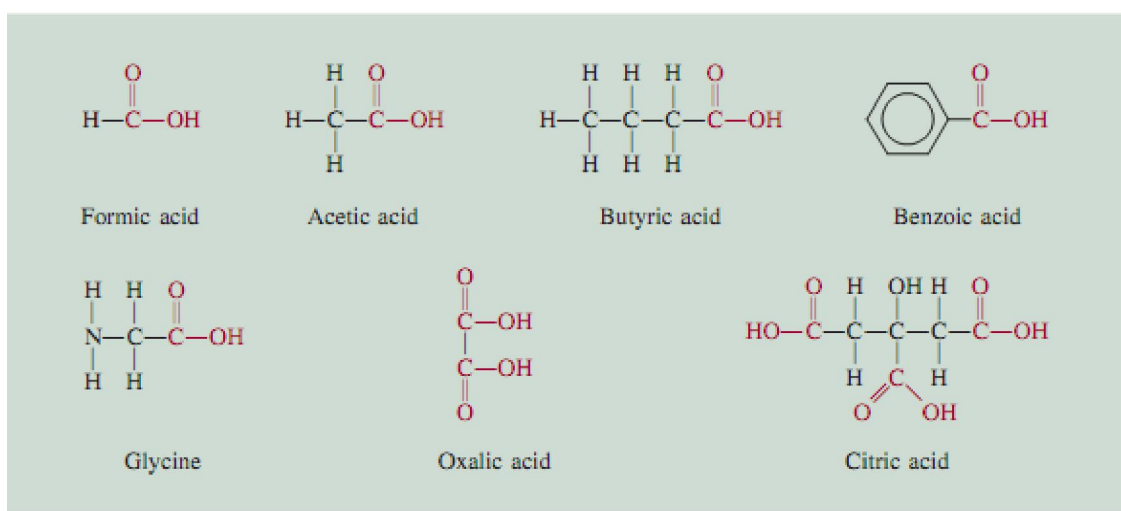
Under appropriate conditions both alcohols and aldehydes can be oxidized to carboxylic

acids, acids that contain the carboxyl group, —COOH :

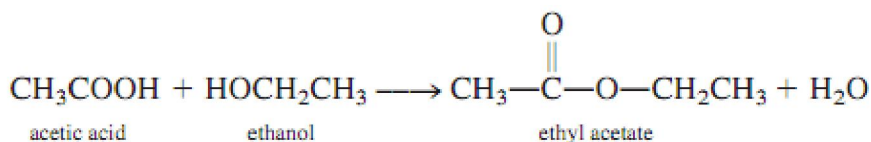


These reactions occur so readily, in fact, that wine must be protected from atmospheric oxygen while in storage. Otherwise, it would soon turn to vinegar due to the formation of acetic acid. The Figure below shows the structure of some of the common carboxylic acids.

Carboxylic acids are widely distributed in nature; they are found in both the plant and animal kingdoms. All protein molecules are made of amino acids, a special kind of carboxylic acid containing an amino group (—NH_2) and a carboxyl group (—COOH).



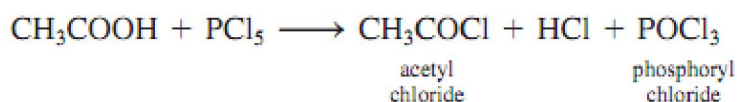
Unlike the inorganic acids HCl , HNO_3 , and H_2SO_4 , carboxylic acids are usually weak. They react with alcohols to form pleasant-smelling esters:



Other common reactions of carboxylic acids are neutralization



and formation of acid halides, such as acetyl chloride



Acid halides are reactive compounds used as intermediates in the preparation of many other organic compounds.

Esters

Esters have the general formula $\text{R}'\text{COOR}$, in which R' can be H, an alkyl, or an aromatic hydrocarbon group and R is an alkyl or an aromatic hydrocarbon group.

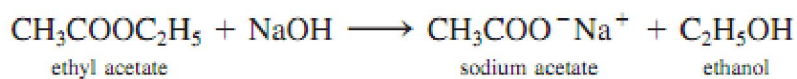
Esters are used in the manufacture of perfumes and as flavoring agents in the confectionery and soft-drink industries. Many fruits owe their characteristic smell and flavor to the presence of esters. For example, bananas contain isopentyl acetate

$[\text{CH}_3\text{COOCH}_2\text{CH}_2\text{CH}(\text{CH}_3)_2]$, oranges contain octyl acetate ($\text{CH}_3\text{COOC}_8\text{H}_{17}$), and apples contain methyl butyrate ($\text{CH}_3\text{CH}_2\text{CH}_2\text{COOCH}_3$).

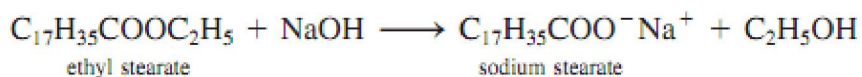
The functional group in esters is $-\text{COOR}$. In the presence of an acid catalyst, such as HCl, esters undergo a reaction with water (a hydrolysis reaction) to regenerate a carboxylic acid and an alcohol. For example, in acid solution, ethyl acetate is converted to acetic acid:



However, this reaction does not go to completion because the reverse reaction, that is, the formation of an ester from an alcohol and an acid, also occurs to an appreciable extent. On the other hand, when the hydrolysis reaction is run in aqueous NaOH solution, ethyl acetate is converted to sodium acetate, which does not react with ethanol, so this reaction goes to completion from left to right:



The term saponification (meaning soap making) was originally used to describe the reaction between an ester and sodium hydroxide to yield soap (sodium stearate):



Saponification is now a general term for alkaline hydrolysis of any type of ester. Soaps are characterized by a long nonpolar hydrocarbon chain and a polar head (the $-\text{COO}^-$ group). The hydrocarbon chain is readily soluble in oily substances, while the ionic carboxylate group ($-\text{COO}^-$) remains outside the oily nonpolar surface. The Figure below shows the action of soap.

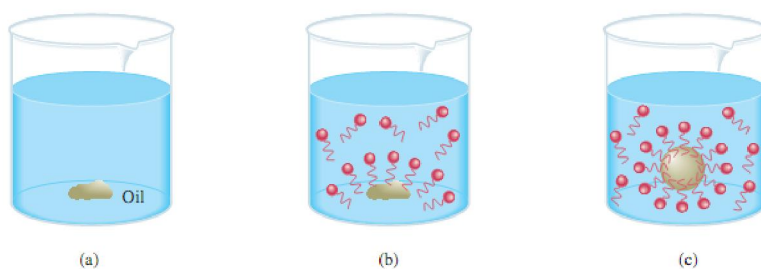


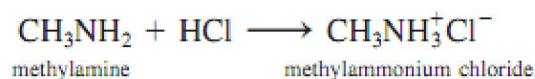
Figure 11.19
The cleansing action of soap. The soap molecule is represented by a polar head and zigzag hydrocarbon tail. An oily spot (a) can be removed by soap (b) because the nonpolar tail dissolves in the oil, and (c) the entire system becomes soluble in water because the exterior portion is now ionic.

Amines

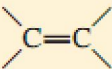
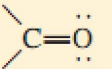
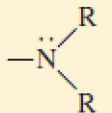
Amines are organic bases that have the general formula R_3N , in which one of the R groups must be an alkyl group or an aromatic hydrocarbon group. Like ammonia, amines are weak Brønsted bases that react with water as follows:



Like all bases, the amines form salts when allowed to react with acids:

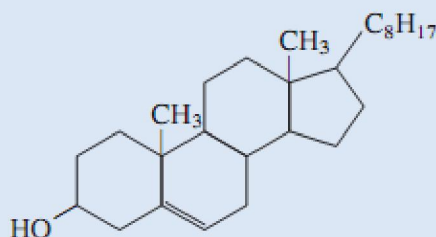


These salts are usually colorless, odorless solids that are soluble in water. Many of the aromatic amines are carcinogenic

Table 11.4 Important Functional Groups and Their Reactions		
Functional Group	Name	Typical Reactions
	Carbon-carbon double bond	Addition reactions with halogens, hydrogen halides, and water; hydrogenation to yield alkanes
$-\text{C}\equiv\text{C}-$	Carbon-carbon triple bond	Addition reactions with halogens, hydrogen halides; hydrogenation to yield alkenes and alkanes
$-\ddot{\text{X}}:$ (X = F, Cl, Br, I)	Halogen	Exchange reactions: $\text{CH}_3\text{CH}_2\text{Br} + \text{KI} \longrightarrow \text{CH}_3\text{CH}_2\text{I} + \text{KBr}$
$-\ddot{\text{O}}-\text{H}$	Hydroxyl	Esterification (formation of an ester) with carboxylic acids; oxidation to aldehydes, ketones, and carboxylic acids
	Carbonyl	Reduction to yield alcohols; oxidation of aldehydes to yield carboxylic acids
$\begin{array}{c} :\text{O}: \\ \\ -\text{C}-\ddot{\text{O}}-\text{H} \end{array}$	Carboxyl	Esterification with alcohols; reaction with phosphorus pentachloride to yield acid chlorides
$\begin{array}{c} :\text{O}: \\ \\ -\text{C}-\ddot{\text{O}}-\text{R} \end{array}$ (R = hydrocarbon)	Ester	Hydrolysis to yield acids and alcohols
 (R = H or hydrocarbon)	Amine	Formation of ammonium salts with acids

Example

Cholesterol is a major component of gallstones, and it is believed that the cholesterol level in the blood is a contributing factor in certain types of heart disease. From the following structure of the compound, predict its reaction with (a) Br_2 , (b) H_2 (in the presence of a Pt catalyst), (c) CH_3COOH .



Strategy To predict the type of reactions a molecule may undergo, we must first identify the functional groups present (see Table 11.4).

Solution There are two functional groups in cholesterol: the hydroxyl group and the carbon-carbon double bond.

- (a) The reaction with bromine results in the addition of bromine to the double-bonded carbons, which become single-bonded.

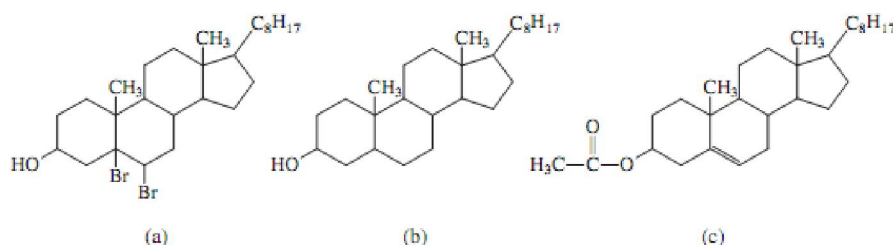


Figure 11.20

The products formed by the reaction of cholesterol with (a) molecular bromine, (b) molecular hydrogen, and (c) acetic acid.

- (b) This is a hydrogenation reaction. Again, the carbon-carbon double bond is converted to a carbon-carbon single bond.
- (c) The acetic acid (CH_3COOH) reacts with the hydroxyl group to form an ester and water. Figure 11.20 shows the products of these reactions.

Practice Exercise Predict the products of the following reaction:



Summary of Facts and Concepts

1. Because carbon atoms can link up with other carbon atoms in straight and branched chains, carbon can form more compounds than most other elements.
2. Alkanes and cycloalkanes are saturated hydrocarbons. Methane, CH_4 , is the simplest of the alkanes, a family of hydrocarbons with the general formula $\text{C}_n\text{H}_{2n+2}$. The cycloalkanes are a subfamily of alkanes whose carbon atoms are joined in a ring. Ethylene, $\text{CH}_2=\text{CH}_2$, is the simplest of alkenes, a class of hydrocarbons

containing carbon-carbon double bonds and having the general formula C_nH_{2n} . Unsymmetrical alkenes can exist as cis and trans isomers. Acetylene, $CH\equiv CH$, is the simplest of the alkynes, which are compounds that have the general formula C_nH_{2n-2} and contain carbon-carbon triple bonds. Compounds that contain one or more benzene rings are called aromatic hydrocarbons. The stability of the benzene molecule is the result of electron delocalization.

3. Functional groups determine the chemical reactivity of molecules in which they are found. Classes of compounds characterized by their functional groups include alcohols, ethers, aldehydes and ketones, carboxylic acids and esters, and amines.

Review Questions

1 Explain why carbon is able to form so many more compounds than most other elements.

2 What is the difference between aliphatic and aromatic hydrocarbons?

3 What do “saturated” and “unsaturated” mean when applied to hydrocarbons? Give examples of a saturated hydrocarbon and an unsaturated hydrocarbon.

4 What are structural isomers?

5 Draw skeletal structures of the boat and chair forms of cyclohexane.

6 What is Markovnikov’s rule?

7 Describe reactions that are characteristic of alkanes, alkenes, and alkynes.

8 Discuss how you can determine which of these compounds might be alkanes, cycloalkanes, alkenes, or alkynes, without drawing their formulas: (a) C_6H_{12} ,

(b) C_4H_6 , (c) C_5H_{12} , (d) C_7H_{14} , (e) C_3H_4

9 Draw the structures of cis-2-butene and trans-2-butene. Which of the two compounds would give off more heat on hydrogenation to butane? Explain.

10 Write the structural formulas for these organic compounds: (a) 3-methylhexane, (b) 1,3,5-trichloro-cyclohexane, (c) 2,3-dimethylpentane, (d) 2-bromo-4-phenylpentane, (e) 3,4,5-trimethyloctane

ANALYSIS OF ORGANIC COMPOUNDS

The following is an outline of the methods used in the study of organic compounds;

PURIFICATION: Before the properties and structure of an organic compound can be completely investigated, the compound must be pure. Common methods of purification are:-

- vii. Recrystallization from suitable solvents.
- viii. Distribution: a) at atmospheric pressure; b) under reduced pressure or in vacuum; c) under increased pressure.
- ix. Steam distribution.
- x. Sublimation.
- xi. Chromatography; and
- xii. Zone melting.

There are various criteria for purity. The most common one for solids is m.p; for liquids b.p; density and refractive index are used; and more recently, the examination of the infra-red spectrum of a compound (whatever its normal physical state) has been used as a test for purity. In all cases, the process of purification is repeated until the physical constant or spectrum remains unchanged.

Before a natural product can be purified, it first has to be extracted from its source. Some of the earlier methods were solvent extraction, distillation and steam distillation. The methods used now are much superior, e.g. counter-current separation (this has replaced solvent extraction), electrophoresis, dialysis, diffusion, molecular distillation, ultracentrifugation, etc. the most important method nowadays is chromatography in its various forms. Zone melting is now also used for separating constituents in a mixture.

QUALITATIVE ANALYSIS

The elements commonly found in organic substances are: Carbon (always: by definition), hydrogen, oxygen, nitrogen, halogens, sulphur, phosphorus and metals.

- i. Carbon and hydrogen. The compound is intimately mixed with dry cupric oxide and the mixture then heated in a tube. Carbon is oxidized to carbon dioxide (detected by means of calcium hydroxide solution), and hydrogen is oxidized to water (detected by condensation on the cooler parts of the tube).
- ii. Nitrogen, halogens and sulphur. These are all detected by the Lassaigne method. The compound is fused with metallic sodium, whereby nitrogen is converted into sodium cyanide, halogen into sodium halide and sulphur into sodium sulphide.

The presence of these sodium salts is then detected by inorganic qualitative methods.

- iii. Phosphorus: The compound is heated with fusion mixture, whereby the phosphorus is converted into metallic phosphate, which is then detected by the usual inorganic tests.
- iv. Metals. When organic compounds containing metals are strongly heated, the organic part usually burns away leaving behind a residue. This residue is usually the oxide of the metal, but in certain cases, it may be the free metal, e.g. silver, or the carbonate, e.g. sodium carbonate.

As a rule, no attempt is made to carry out any test for oxygen; its presence is usually inferred from the chemical properties of the compound.

QUANTITATIVE ANALYSIS

The methods used in the determination of the composition by weight of an organic compound are based on simple principles.

- i. Carbon and hydrogen are estimated by burning a known weight of the substance in a current of oxygen, and weighing the carbon dioxide and water formed. If elements (non-metallic) other than carbon, hydrogen and oxygen are present, special precautions must be taken to prevent their interfering with the estimation of the carbon and hydrogen.
- ii. Nitrogen may be estimated in several ways, but only two are commonly used.
 - a) Dumas' method. This consists in oxidizing the compound with copper oxide, and measuring the volume of nitrogen found. This method is applicable to all organic compounds containing nitrogen.
 - b) Kjeldahl's method. This depends on the fact that when organic compounds containing nitrogen are heated with concentrated sulphuric acid, the organic nitrogen is converted into ammonium sulphate. This method however, has certain limitations.
- iii. Oxygen is usually estimated by difference. There are, however, some semi-micro methods e.g. the organic compound is subjected to pyrolysis in a stream of nitrogen, and all oxygen in the pyrolysis products is converted into carbon monoxide by passage over carbon heated at 1120°C . The carbon monoxide is then passed over iodine pentoxide, and the iodine liberated is estimated titrimetrically.
- iv. Chlorine, bromine, iodine, sulphur and phosphorus are usually estimated by the 'oxygen flask' method of combustion. A sample of the compound is wrapped in ashless filter paper and ignited electrically in a flask of oxygen containing the appropriate absorption liquid for the halogen or sulphur oxides produced.

Suitable titration (or gravimetric determination) gives the amounts of these elements present. For example, sulphur oxides are absorbed in hydrogen peroxide and the sulphuric acid produced titrated with standard alkali.

Fluorine is estimated by fusing with sodium or potassium in a nickel bomb, and the alkali, fluorine is then determined gravimetrically or by titration.

Quantitative analysis falls into three groups according to the amount of material used for the estimation:

- i. Macro-methods which require about 0.1 – 0.5g of material (actual amount depends on the element being estimated).
- ii. Semi-micro methods which require 20 – 50mg of material.
- iii. Micro-methods which require 3 – 5mg of material.

The micro-methods are almost always used now. Furthermore, with the increasing use of superior methods of separation, there is a trend to use still smaller, amounts, the range being from 500 to 30mg, and there is an increasing tendency to use instrumental techniques.

EMPIRICAL FORMULS DETERMINATION

The empirical formula indicates the relative numbers of each kind of atom in the molecule, and is calculated from the percentage composition of the compound. The following example illustrates the procedure.

- 1) 0.202g of a substance gives on combustion 0.361g of CO₂ and 0.147g of H₂O. What is the empirical formula of the substance?

$$\text{Wt. of C in sample} = \frac{12}{44} \times 0.361 = 0.0985\text{g}$$

$$\text{Wt. of H in sample} = \frac{2}{18} \times 0.147 = 0.0163\text{g}$$

$$\text{Wt. of O} = 0.202 - (0.0985 + 0.0163) = 0.0872\text{g.}$$

- i. Percentage composition

$$\% \text{ C} = \frac{0.0985}{0.202} \times 100 = 48.76$$

$$\% \text{ H} = \frac{0.0163}{0.202} \times 100 = 8.07$$

In practice, the percentages of C and H are evaluated and the percentage of O is obtained by subtraction of their sum from 100, i.e.

$$\% \text{ O} = 100 - (48.76 + 8.07) = 43.17$$

ii. Empirical formula

- a) If given the weights of simple and combustion products, divide the weights of C, H and O (by difference) their respective atomic weights. Thus, the ratio of atoms is

$$\text{C: H: O} = \frac{0.0985}{12} : \frac{0.0163}{1} : \frac{0.0872}{16}$$

To obtain this ratio as whole numbers, divide throughout by the smallest number.

$$\text{C: H: O} = 0.0082 : 0.00163 : 0.00545$$

$$= \text{C: H: O} = 1.50 : 2.99 : 1$$

(Had these numbers been integers or very near to integers e.g. 1.02: 3.04: 1 then, the empirical formula would have been CH_3O).

The procedure is to multiply the ratio by small numbers until a ratio of nearly whole numbers is obtained. Thus, multiplying by 2 give the ratio 3.00: 5.98: 2, the empirical formula is therefore, $\text{C}_3\text{H}_6\text{O}_2$.

- b) If given the percentage composition, the procedure is similar to the above i.e.

$$\text{C: H: O} = \frac{48.76}{12} : \frac{8.07}{1} : \frac{43.17}{16}$$

$$= 4.06 : 8.07 : 2.7$$

$$= 1.50 : 2.99 : 1$$

Multiply by 2 gives 3: 5.98: 2 and so the empirical formula is $\text{C}_3\text{H}_6\text{O}_2$.

It should be noted that the final ratio is the same as before. This is to be expected since, to obtain the percentage composition, each weight was multiplied by the same constant $\frac{100}{0.202}$.

MOLECULAR WEIGHT DETERMINATION

The molecular formula: This gives the actual number of atoms of each kind in the molecule – is obtained by multiplying the empirical formula by some whole number which is obtained from consideration of the molecular weight of the compound. In many cases, this whole number is one.

The methods used for the determination of molecular weights are mainly physical.

The standard physical methods are the determination of:

- i) vapour density
 - ii) elevation of boiling point
 - iii) depression of freezing point.
- These methods are described fully in physical chemistry. In addition to these standard methods, which are used mainly for relatively simple molecules, there are other physical methods used for compounds having high molecular weights, e.g. rate of diffusion, rate of sedimentation, viscosity of the solution, osmotic pressure. X-ray analysis and mass spectrometry are also used for molecular weight determination.

DETERMINATION OF STRUCTURE.

Determination of structure, i.e., the manner in which the atoms are arranged in the molecule. The usual procedure for elucidating the structure of an unknown compound is to make a detailed study of its chemical reactions. This procedure is known as the analytical

methods and includes breaking down (degrading) the compound into smaller molecules of known structure.

When sufficient evidence has been accumulated, a tentative structure which best fits the facts is accepted. Sometimes two (or even more) structures fit the facts almost equally well, and it has been shown in certain cases that the compound exists in both forms which are in equilibrium. This phenomenon is known as tautomerism. Where tautomerism has not been shown to be present one must accept (with reserve) the structure that has been chosen.

Until recently, structure determinations were based almost completely on purely chemical evidence. Nowadays, physical methods are considered as necessary tools for elucidating structures. They are used on the compound itself, and may also be used in the examination of the fragments obtained by degradative work. Of all the physical methods used, infra-red spectroscopy is probably the most widely applicable.

Stereochemistry

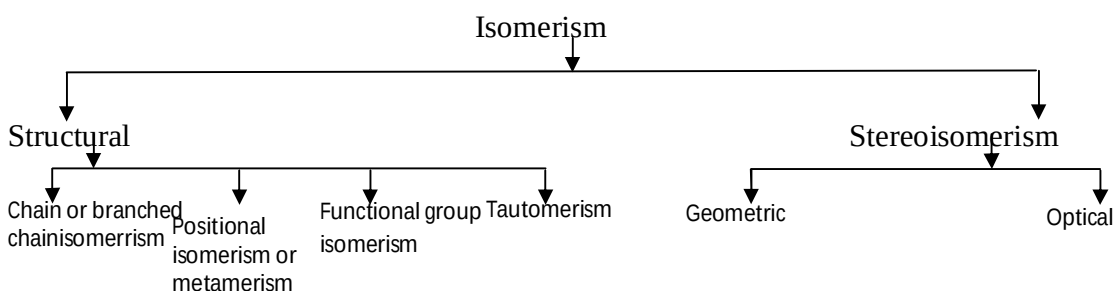
The science of organic chemistry is based on the relationship between molecular structure and properties. That part of the science which deals with structure in three dimensions is called stereochemistry.

Isomerism

Isomerism is the existence of two or more compounds with the same molecular formula but different molecular structures due to the difference in the arrangement of atoms or spatial orientation of atoms. Hence such compounds differ in physical and/or chemical properties. The possible structures are known as isomers.

On a broad basis, isomerism is categorized into two types:

- Structural (Constitutional) isomerism and
- stereoisomerism.



Structural isomerism arises as a result of difference in the arrangement of atoms of a molecular formula.

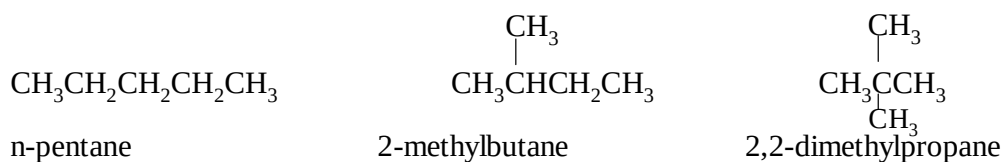
(i) Chain isomerism or branched chain isomerism

These are isomers having the same number and kind of atoms but having different structures due to the presence of branched chains. E.g. butane has two isomers, pentane has three, hexane has five and heptane has eight.

For butane

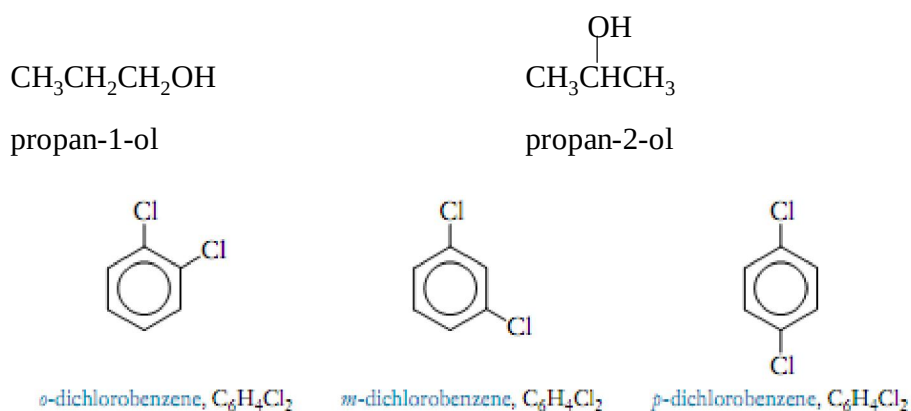


For pentane



(ii) Metamerism(Positional isomerism)

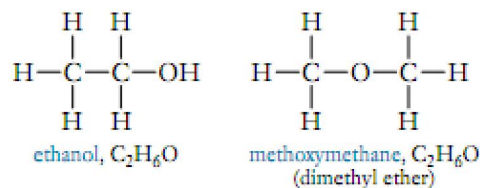
They are isomers which have substituents in different positions on the same carbon skeleton. It occurs when large alkyl groups are attached to the same functional groups as in alkanol, alkylhalides or ether. For example, there are two structural isomeric alcohols corresponding to the formula $\text{C}_3\text{H}_7\text{OH}$ and three for dichlorobenzene.



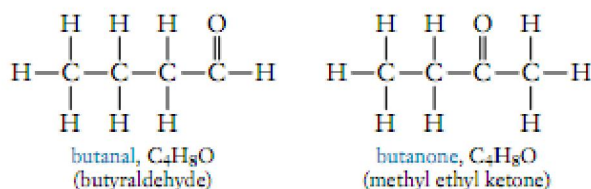
(iii) Functional Group isomerism

This is an isomerism between compounds in different homologous series. For example, there are two isomers corresponding to the formula $\text{C}_2\text{H}_6\text{O}$

An alcohol and an ether:

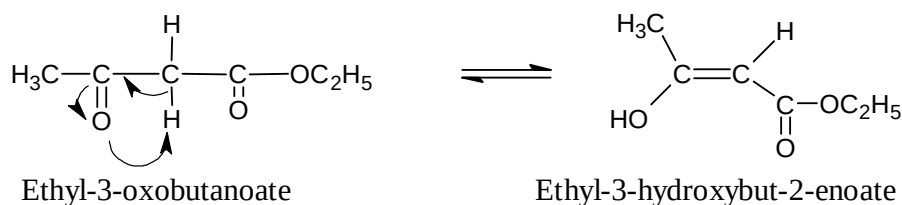


An aldehyde and a ketone:



(iv) Tautomerism

It is a special type of isomerism in which isomers exist together in dynamic equilibrium



Stereoisomerism

Another aspect of stereochemistry is stereoisomerism. Isomers are different compounds that have the same molecular formula. The particular kind of isomers that are different from each other only in the way the atoms are oriented in space (but are like one another with respect to which atoms are joined to which other atoms) are called stereoisomers.

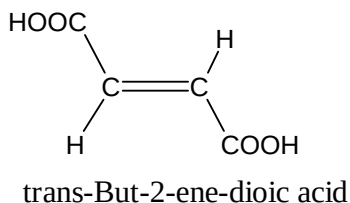
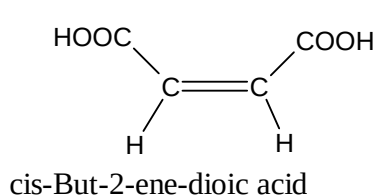
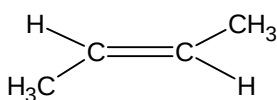
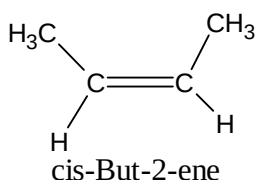
There are two types of stereoisomers: geometric and optical

Geometric Isomerism

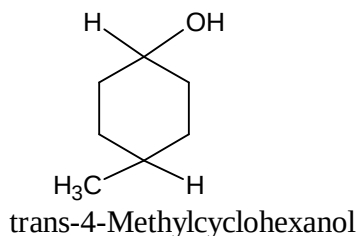
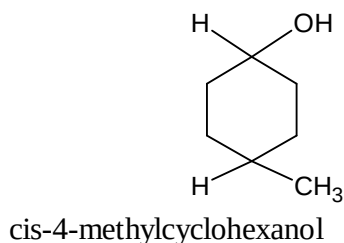
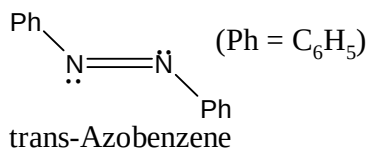
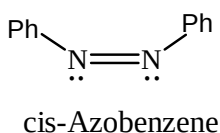
Geometric isomers (or cis-trans isomers) differ only in the spatial orientation of groups about a plane or direction. Two geometric isomers have the same molecular formula, the same functional groups, the same base chain or ring, and the same order of attachment of atoms; they differ in orientation either (1) around a double bond or (2) across the ring in a cyclic compound. If the larger groups are on opposite sides of the ring or the double bond, the designation trans appears in the name; if they are on the same side, the

designation is cis. The cis- and trans- isomers are not mirror images of each other and are called diastereomers.

For example, cis-1,2-dibromoethene and trans-1,2-dibromoethene are isomers, since they have the same molecular formula $C_2H_2Br_2$. The difference comes not from the connectivity of their atoms but from their arrangement in space. Both molecules have two central carbons, linked by a double bond, and have each carbon connected to one bromine and one hydrogen atom. However, in cis-1,2-dibromoethene, the bromine atoms are positioned on the same side of the double bond, while in trans-1,2-dibromoethene, they are on opposite sides.

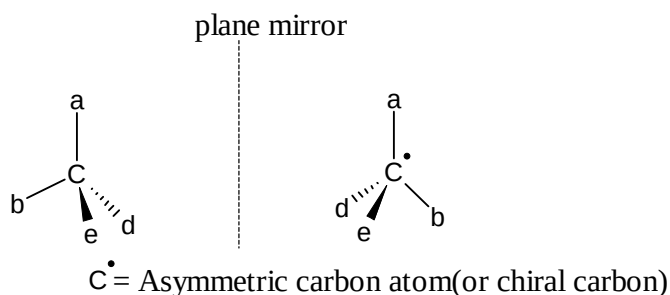


The geometry of molecules containing a nitrogen-nitrogen double bond or carbon-nitrogen double bond makes them also to be capable of cis-trans isomerism, for example in azobenzene



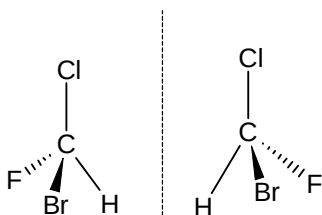
Optical isomerism

Optical isomerism is observed in compounds in which four different groups are attached to the same carbon atom. There are two isomers that result from this type of isomerism. These isomers are not superimposable on their mirror images, and are referred to as enantiomers, also known as enantiomorphs, optical antipodes, chiral molecules or optical isomers. The central atom, to which the four different atoms or groups are attached is called a chiral centre. If the central atom is carbon, it is called Asymmetric carbon atom or chiral carbon.

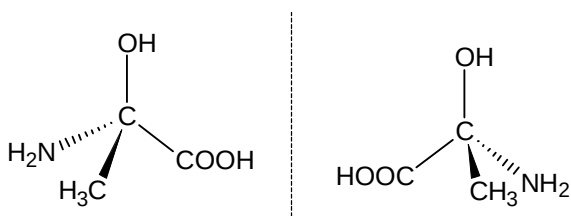


Examples include:

(a) Optical isomers of Bromochlorofluoromethane:



(b) Optical isomers of Aminopropanoic acid (Alanine):



Stereoisomers have the same type and number of atoms, connected in the same order, but arranged differently in space. Optical isomers (enantiomers) and geometric isomers are subgroups of stereoisomers. They differ, however, in that geometric isomers have different physical and chemical properties, whereas optical isomers have physical properties that are identical (e.g., melting point, boiling point, and density).

Optical isomers also undergo the same chemical reactions, except when they interact with other chiral compounds. Consequently, their properties in biological systems may be very different. They also often exhibit different solubilities in solvents that are composed of chiral molecules.

Optical isomers also differ from each other in one notable physical property: They interact with polarized light in different ways. One of the optical isomers is designated as the D-isomer (DEXTROROTATORY i.e. rotation of the plane in a clockwise direction) and its nonsuperimposable mirror image is designated as the L-isomer (LAEVOROTATORY i.e rotation of the plane in an anticlockwise direction). The phenomenon in which a plane of polarized light is rotated by samples of either isomer is called optical activity. It can be measured with a polarimeter. A racemic mixture is a single sample containing equal amounts of the two optical isomers of a compound. Such a solution does not rotate a plane of polarized light because the equal and opposite effects of the two isomers exactly cancel. The isomers must be separated from each other to exhibit optical activity. One very important way in which optical isomers differ chemically from one another is in their biological activities.

Organic molecules are based on a framework of carbon–carbon and carbon–hydrogen bonds. Many compounds contain only the two elements C and H; they are called hydrocarbons.

Chemistry of Macromolecules

Macromolecules are large molecules, most of which are carbon compounds. Naturally occurring compounds of this type are of great significance in nature. Macromolecules include carbohydrates, lipids(fats and oils), proteins and nucleic acids. Many synthetic compounds of industrial importance are macromolecules including the plastics, both thermoplastics such as polythene, terylene, nylon.e.t.c and thermosetting materials such as bakelite, melamine and some polyester resins.

Carbohydrates

Carbohydrates have the general formula $C_n (H_2O)_m$, thus the name hydrated carbons or, more commonly, carbohydrates. Monosaccharides, or simple sugars, are the monomers from which more complex carbohydrates are constructed. The most important simple sugars contain either five or six carbon atoms. Monosaccharides that contain an aldehyde group are called aldoses, and those that contain a ketone group are called ketoses. The most commonly encountered monosaccharide is glucose, $C_6H_{12}O_6$, also known as

dextrose or blood sugar. The figure below shows glucose as it is often written to emphasize its aldehyde functional group.

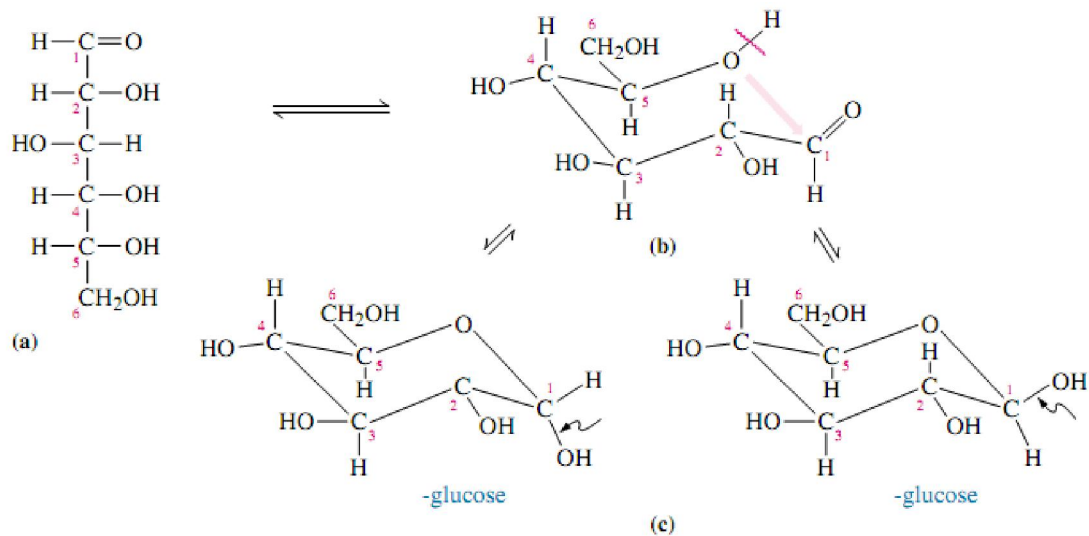
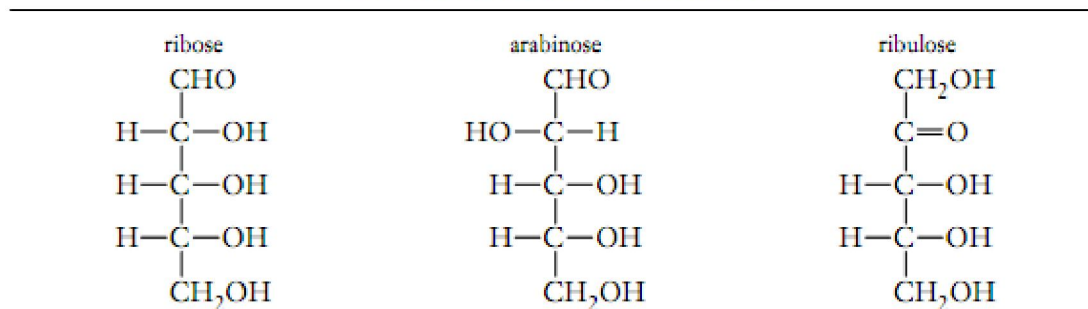


Figure showing the cyclization of glucose to form either the α - or the β - cyclic form. (a) The linear representation of the aldehyde form of glucose. (b) A coiled representation that shows the aldehyde function group near the —OH group of carbon 5. (c) The two cyclic forms of glucose.

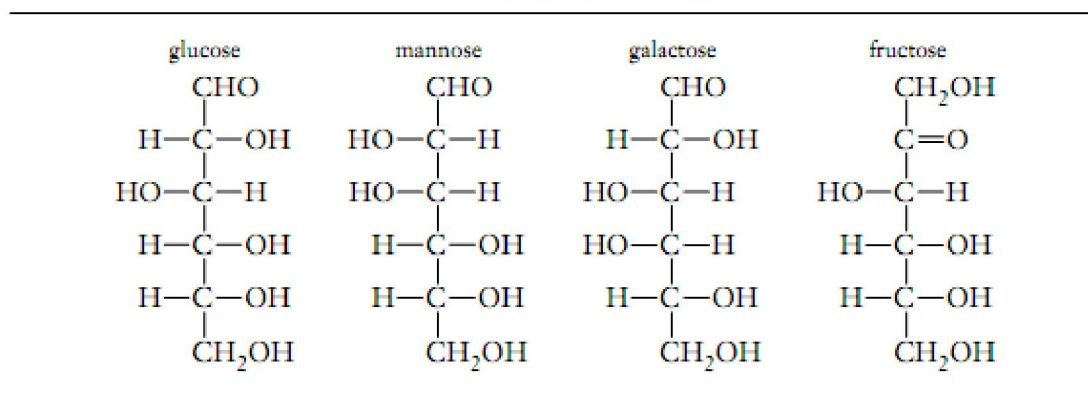
We can see from this formula that carbons 2, 3, 4, and 5 in glucose each have four different groups and are, therefore, chiral. Each different monosaccharide has a unique arrangement of atoms about its chiral atoms. Some important monosaccharides are shown in the Table below.

Some Important Monosaccharides

Five Carbon Atom Monosaccharides



Six Carbon Atom Monosaccharides



Besides their aldehyde or ketone form, saccharides can also exist in a cyclic form. The figure above shows the interconversion of glucose between its linear form and its cyclic form. Figure b shows the molecule rotated into a conformation that brings the hydroxyl group of carbon 5 near the aldehyde carbon. In this conformation, a reversible reaction occurs in which a hydrogen is moved from the hydroxyl group to the carbonyl oxygen, and an ether bond is formed between the two carbons (Figure c). During the formation of the ether bond, a ring is also formed and carbon atom 1 becomes chiral. This new chiral center can exist as either of two isomers; the resulting stereoisomers are referred to as α - or β -forms.

Most naturally occurring carbohydrates contain more than one monosaccharide unit. Complex carbohydrates are formed when two or more monosaccharides are linked. This linkage between two monosaccharides occurs by elimination of a molecule of water and the formation of an ether bond between two of the cyclic structures, as shown in Figure below. This illustrates the linkage of α -glucose and β -fructose to form sucrose. The newly formed C—O bond linking the rings is called the glycosidic bond.

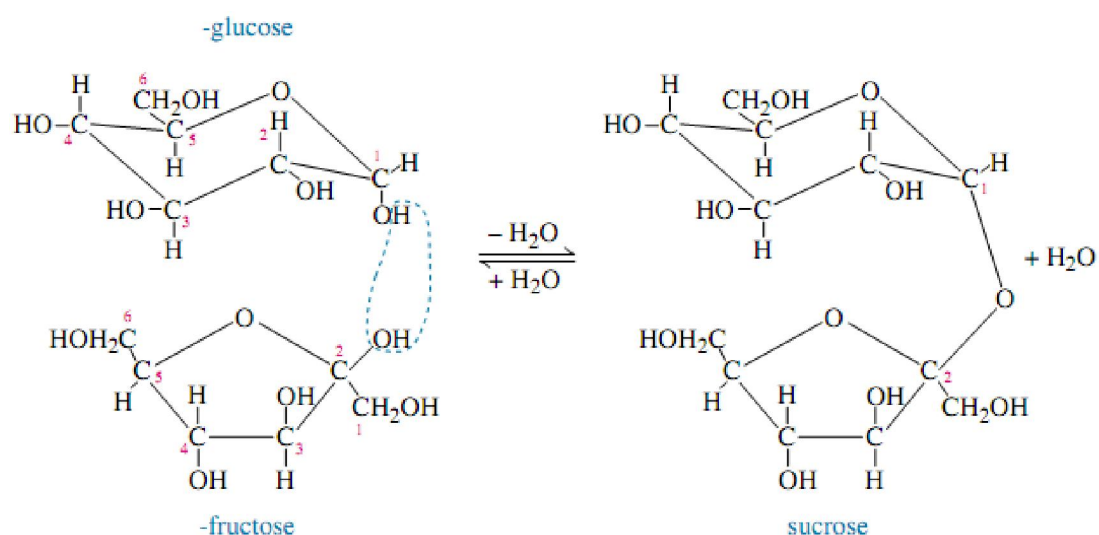
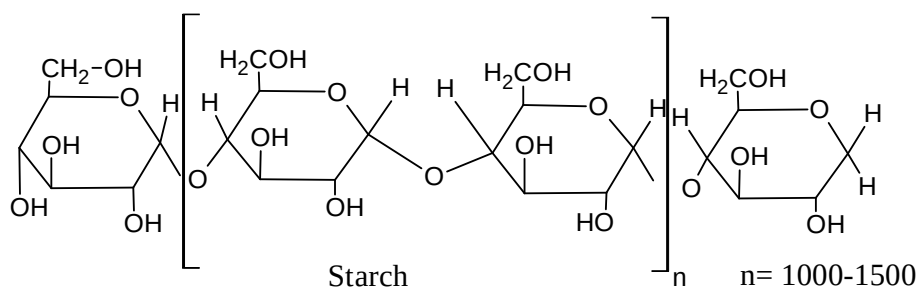


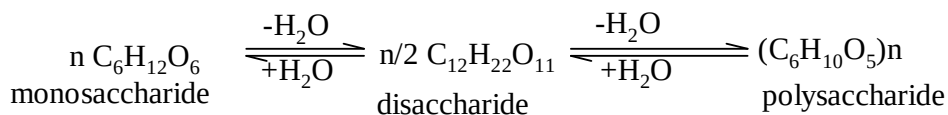
Figure showing the Formation of sucrose by the condensation of an α -glucose molecule with a β -fructose molecule.

Fructose is a ketone-containing carbohydrate that exists in the cyclic form shown. A disaccharide is a molecule consisting of two monosaccharides joined together by a glycosidic bond. Sucrose is a disaccharide; two other disaccharides, each involving only glucose units, In such a molecule, the C—O—C linkage between the monosaccharide units involves carbon number 1 or 2 of one monosaccharide and carbon number 1, 4, or 6 of the other monosaccharide.

A carbohydrate that contains three monosaccharides is called a trisaccharide. Carbohydrates that contain from four to ten monosaccharides are usually called by the general term oligosaccharides. Carbohydrates that contain larger numbers of monosaccharide units are called polysaccharides.e.g. cellulose and starch



Starch is a polysaccharide with a relative molecular mass of 15000 to 300000. It is composed of thousands of repeating glucose units linked together to form giant branched molecule. Starch consists of α (+) glucose units linked by the C-1 and C-4 atoms. The relationship between mono-, di- and polysaccharides is



Reactions of starch

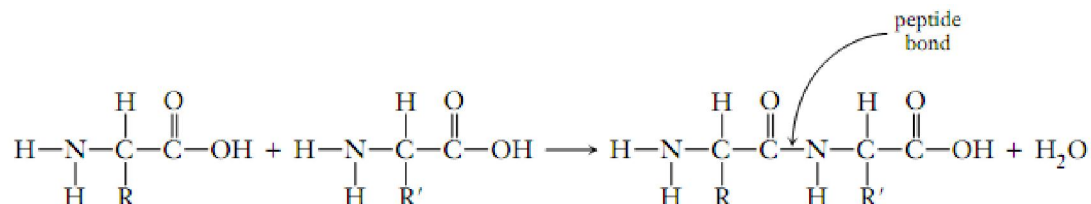
Starch does not reduce Fehling's solution, Tollen's reagent or ammoniacal silver nitrate, nor does it form an osazone with phenylhydrazine.

Starch forms an intense dark blue substance with iodine, and the color disappears on warming and reappears on cooling, and thus provides a sensitive test for either iodine or starch.

Starch occurs naturally in most plants as a storage carbohydrate, e.g. maize (corn), cassava, yam and cocoyam tuber, potatoes, rice, wheat, barley and other cereals.

Polypeptides and Proteins

Two amino acids can react to form an amide bond. Compounds formed by the linking of small numbers of amino acids are called peptides, and the amide bond linking the amino acid monomers is called the peptide bond.



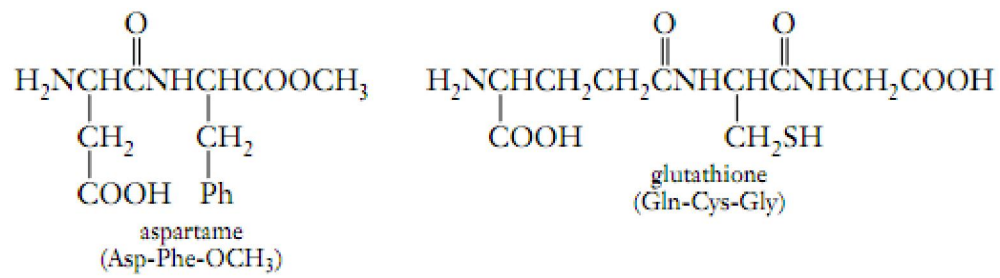
Prefixes di-, tri-, and so on are used to indicate the number of amino acid monomers that are joined to form the peptide. The compound indicated in the preceding formula is a dipeptide.

Twenty different α -amino acids are found widely in nature. Each amino acid is commonly given a three-letter abbreviation. These acids differ in the R groups attached to the α -carbon (the carbon adjacent to the —COOH group). Differences in R groups lead to differences in the properties of the α -amino acids. The amino acids are usually classified by their R groups, or side chains, according to two criteria. The first classification depends on whether the R groups are polar or nonpolar, the second describes whether the R group contains an acidic or basic group. The 20 common amino acids are shown, according to these classifications, in the Table below.

The 20 Common Amino Acids Found in Proteins

Nonpolar R Groups			
alanine (Ala)	$\text{CH}_3\text{CH}(\text{NH}_2)\text{COOH}$	phenylalanine (Phe)	$\text{C}_6\text{H}_5\text{CH}_2\text{CH}(\text{NH}_2)\text{COOH}$
glycine (Gly)	$\text{HCH}(\text{NH}_2)\text{COOH}$	proline (Pro)	$\text{C}_4\text{H}_7\text{NO}_2$
isoleucine (Ile)	$\text{CH}_3\text{CH}_2\text{CH}(\text{CH}_3)\text{CH}(\text{NH}_2)\text{COOH}$	tryptophan (Trp)	$\text{C}_8\text{H}_7\text{NO}_2$
leucine (Leu)	$(\text{CH}_3)_2\text{CHCH}_2\text{CH}(\text{NH}_2)\text{COOH}$	valine (Val)	$(\text{CH}_3)_2\text{CHCH}(\text{NH}_2)\text{COOH}$
methionine (Met)	$\text{CH}_3\text{SCH}_2\text{CH}_2\text{CH}(\text{NH}_2)\text{COOH}$		
Polar, Neutral R Groups			
asparagine (Asn)	$\text{H}_2\text{NC}(=\text{O})\text{CH}_2\text{CH}(\text{NH}_2)\text{COOH}$	serine (Ser)	$\text{HOCH}_2\text{CH}(\text{NH}_2)\text{COOH}$
glutamine (Gln)	$\text{H}_2\text{NC}(=\text{O})\text{CH}_2\text{CH}_2\text{CH}(\text{NH}_2)\text{COOH}$	threonine (Thr)	$\text{CH}_3\text{CH}(\text{OH})\text{CH}(\text{NH}_2)\text{COOH}$
Polar, Acidic R Groups		Polar, Basic R Groups	
aspartic acid (Asp)	$\text{HOOCCH}_2\text{CH}(\text{NH}_2)\text{COOH}$	arginine (Arg)	$\text{H}_2\text{NC}(=\text{NH})\text{NHCH}_2\text{CH}_2\text{CH}_2\text{CH}(\text{NH}_2)\text{COOH}$
glutamic acid (Glu)	$\text{HOOCCH}_2\text{CH}_2\text{CH}(\text{NH}_2)\text{COOH}$	histidine (His)	$\text{C}_6\text{H}_7\text{NO}_2$
cysteine (Cys)	$\text{HSCH}_2\text{CH}(\text{NH}_2)\text{COOH}$	lysine (Lys)	$\text{H}_2\text{NCH}_2\text{CH}_2\text{CH}_2\text{CH}_2\text{CH}(\text{NH}_2)\text{COOH}$
tyrosine (Tyr)	$\text{HO}-\text{C}_6\text{H}_4-\text{CH}_2\text{CH}(\text{NH}_2)\text{COOH}$		

Many small peptides are known to have physiological significance. Aspartame, a common artificial sweetener, is a dipeptide. Glutathione, a tripeptide, acts as a scavenging agent for harmful oxidizing agents believed to cause cancer. Enkephalins, which are naturally occurring analgesics (pain relievers) that occur in the brain, are pentapeptides. The antibiotics gramicidin-S and tyrocidine A are cyclic decapeptides (i.e., they contain ten amino acids linked into a ring by peptide bonds) produced by a bacterium.



A protein is a biopolymer, of molecular weight typically 5000 g/mol or greater, consisting of one or more polypeptide chains. The α -amino acids that occur in nearly all proteins are the L optical isomer. Proteins make up more than 50% of the dry weight of animals and bacteria. They perform many important functions in living organisms, a few of which are indicated in the Table below. Each protein carries out a specific biochemical function. Each is a polypeptide with its own unique sequence of amino acids. The amino acid sequence of a protein determines exactly how it folds up in a three-dimensional conformation and how it performs its precise biochemical task.

Some Functions of Proteins

Example	Function
Enzymes	
amylase	converts starch to glucose
DNA polymerase I	repairs DNA molecule
trans aminase	transfers amino group from one amino acid to another
Structural Proteins	
viral coat proteins	outer covering of virus
keratin	hair, nails, horns, hoofs
collagen	tendons, cartilage
Hormones	
insulin, glucagon	regulate glucose metabolism
oxytocin	regulates milk production in female mammals
vasopressin	increases retention of water by kidney
Contractile Proteins	
actin	thin contractile filaments in muscle
myosin	thick filaments in muscle
Storage Proteins	
casein	a nutrient protein in milk
ferritin	stores iron in spleen and egg yolk
Transport Proteins	
hemoglobin	carries O ₂ in blood
myoglobin	carries O ₂ in muscle
serum albumin	carries fatty acids in blood
cytochrome c	transfers electrons
Immunological Proteins	
γ-globulins	form complexes with foreign proteins
Toxins	
neurotoxin	blocker of nerve function in cobra venom
ricin	nerve toxin in South American frog (most toxic substance known—0.000005 g is fatal to humans)

An enzyme is a protein that catalyzes a biological reaction. Each protein consists of one or more polypeptide chains with a unique sequence of amino acids. The order of these monomers is termed the primary structure of the protein. The many possible interactions between different amino acids in the polypeptide chain cause each protein molecule to adopt a characteristic shape, usually depending on its environment. The secondary structure of a protein is the arrangement in space of the polypeptide backbone, without reference to the conformations of the side chains (R groups).

Myoglobin is the oxygen-storing protein in muscles. Its amino acid sequence (primary structure) are shown in the Figure below. The tertiary structure of a protein describes the overall shape of the protein, including the side chains and any other nonpeptide components of the protein. Myoglobin is referred to as a globular protein. Some proteins consist of multiple polypeptide units called subunits. These subunits are held to one another by noncovalent interactions such as hydrogen bonding, dipole–dipole interactions, or ionic attractions, to form the quaternary structure of the protein.

Val-Leu-Ser-Glu-Gly-Glu-Trp-Gln-Leu-Val-Leu-His-Val-Trp-Ala-Lys-
Val-
Glu-Ala-Asp-Val-Ala-Gly-His-Gly-Gln-Asp-Ile-Leu-Ile-Arg-Leu-Phe-
Lys-
Ser-His-Pro-Glu-Thr-Leu-Glu-Lys-Phe-Asp-Arg-Phe-Lys-His-Leu-Lys-
Thr-Glu-Ala-Glu-Met-Lys-Ala-Ser-Glu-Asp-Leu-Lys-Lys-His-Gly-Val-
Thr-
Val-Leu-Thr-Ala-Leu-Gly-Ala-Ile-Leu-Lys-Lys-Lys-Gly-His-His-Glu-
Ala-
Glu-Leu-Lys-Pro-Leu-Ala-Gln-Ser-His-Ala-Thr-Lys-His-Lys-Ile-Pro-Ile-
Lys-
Tyr-Leu-Glu-Phe-Ile-Ser-Glu-Ala-Ile-Ile-His-Val-Leu-His-Ser-Arg-His-
Pro-Gly-Asn-Phe-Gly-Ala-Asp-Ala-Gln-Gly-Ala-Met-Asn-Lys-Ala-Leu-
Glu-Leu-Phe-Arg-Lys-Asp-Ile-Ala-Ala-Lys-Tyr-Lys-Glu-Leu-Gly-Tyr-
Gln-Gly

Synthetic Polymers

A polymer is a large molecule that is a high-molecular-weight chain of small molecules. The small molecules that are linked to form polymers are called monomers. Typical polymers consist of hundreds or thousands of monomer units and have molecular weights up to thousands or millions.

Polymerization is the combination of many small molecules (monomers) to form large molecules (polymers). Polymers are divided into two classes:

Natural and

Synthetic.

Important biological molecules such as proteins, nucleic acids, and polysaccharides (starches and the cellulose in wood and cotton) are natural polymers. Natural rubber and natural fibers such as silk and wool are also natural polymers.

Most synthetic polymers are organic compounds. Familiar examples include plastics such as polyethylene, Teflon, and Lucite (Plexiglas) and synthetic fibers such as nylon, Orlon, and Dacron.

Synthetic Organic Polymers

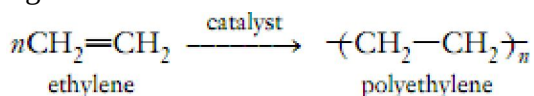
Because of their size, we might expect molecules containing thousands of carbon and hydrogen atoms to form an enormous number of structural and geometric isomers (if C=C bonds are present). However, these molecules are made up of monomers, simple repeating units, and this type of composition severely restricts the number of possible isomers.

Synthetic polymers are created by joining monomers together, one at a time, by means of addition reactions and condensation reactions.

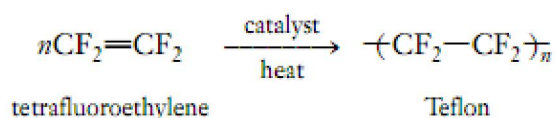
Addition reactions

Addition reactions involve unsaturated compounds containing double or triple bonds, particularly C=C and C≡C. Hydrogenation and reactions of hydrogen halides and halogens with alkenes and alkynes are examples of addition reactions.

Polymerization is an important addition reaction of the alkenes. Polymers formed by this kind of reaction are called addition polymers. The formation of polyethylene is an important example. In the presence of appropriate catalysts (a mixture of aluminum trialkyls, R₃Al, and titanium tetrachloride, TiCl₄), ethylene polymerizes into chains containing 800 or more carbon atoms.

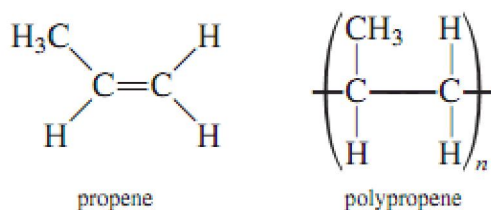


The polymer may be represented as CH₃(CH₂—CH₂)_nCH₃, where n is approximately 400. Polyethylene is a tough, flexible plastic. It is widely used as an electrical insulator and for the fabrication of such items as unbreakable refrigerator dishes, plastic cups, and squeeze bottles. Polypropylene is made by polymerizing propylene, CH₃—CH=CH₂, in much the same way. Teflon is made by polymerizing tetrafluoroethylene in a similar reaction.



The molecular weight of Teflon is about 2 × 10⁶. Approximately 20,000 CF₂=CF₂ molecules polymerize to form a single giant molecule. Teflon is a very useful polymer. It does not react with concentrated acids and bases or with most oxidizing agents. It does not dissolve in most organic solvents.

The chemistry of polymers is more complex if the starting units are asymmetric:

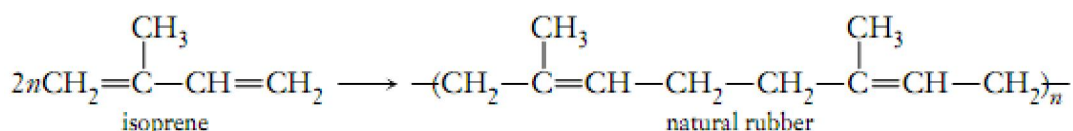


Several geometric isomers can result from an addition reaction of propenes. If the additions occur randomly, we obtain atactic polypropenes, which do not pack together

well. These polymers are rubbery, amorphous, and relatively weak. Two other possibilities are an isotactic structure, in which the R groups are all on the same side of the asymmetric carbon atoms, and a syndiotactic form, in which the R groups alternate on the left and right of the asymmetric carbons. Of these, the isotactic isomer has the highest melting point and greatest crystallinity and is endowed with superior mechanical properties.

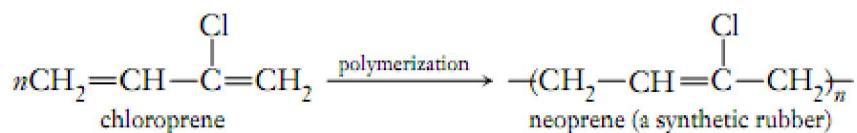
Natural rubber is obtained from the sap of the rubber tree, a sticky liquid called latex.

Rubber is a polymeric hydrocarbon formed in the sap by the combination of about 2000 molecules of 2-methyl-1,3-butadiene, commonly called isoprene. The molecular weight of rubber is about 136,000.



When natural rubber is warmed, it flows and becomes sticky. To eliminate this problem, vulcanization is used. This is a process in which sulfur is added to rubber and the mixture is heated to approximately 140°C. Sulfur atoms combine with some of the double bonds in the linear polymer molecules to form bridges that bond one rubber molecule to another. This cross-linking by sulfur atoms converts the linear polymer into a three-dimensional polymer. Fillers and reinforcing agents are added during the mixing process to increase the durability of rubber and to form colored rubber. Carbon black is the most common reinforcing agent. Zinc oxide, barium sulfate, titanium dioxide, and antimony(V) sulfide are common fillers.

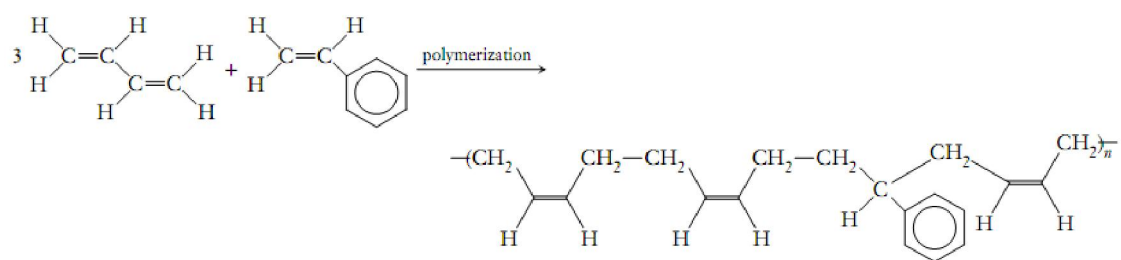
Some synthetic rubbers are superior to natural rubber in some ways. Neoprene is a synthetic elastomer (an elastic polymer) with properties quite similar to those of natural rubber. The basic structural unit is 2-chloro-1,3-butadiene, commonly called chloroprene, which differs from isoprene in having a chlorine atom rather than a methyl group at carbon 2 of the 1,3-butadiene chain.



Neoprene is less affected by gasoline and oil and is more elastic than natural rubber. It resists abrasion well and is not swollen or dissolved by hydrocarbons. It is widely used to make hoses for oil and gasoline, electrical insulation, and automobile and refrigerator parts.

When two different monomers are mixed and then polymerized, copolymers are formed. Depending on the ratio of the two monomers and the reaction conditions, the order of the units can range from quite regular (e.g., alternating) to completely random.

In this way, polymers with a wide variety of properties can be produced. SBR is an example of copolymer. It is formed by the addition of butadiene to styrene in a 3:1 ratio to give styrenebutadiene rubber (SBR). Styrene and butadiene are different monomers,



The double bonds in SBR can be cross-linked by vulcanization as described for natural rubber. SBR is used primarily for making tires. Other copolymers are used to make car bumpers, body and chassis parts, wire insulation, sporting goods, sealants, and caulking compounds.

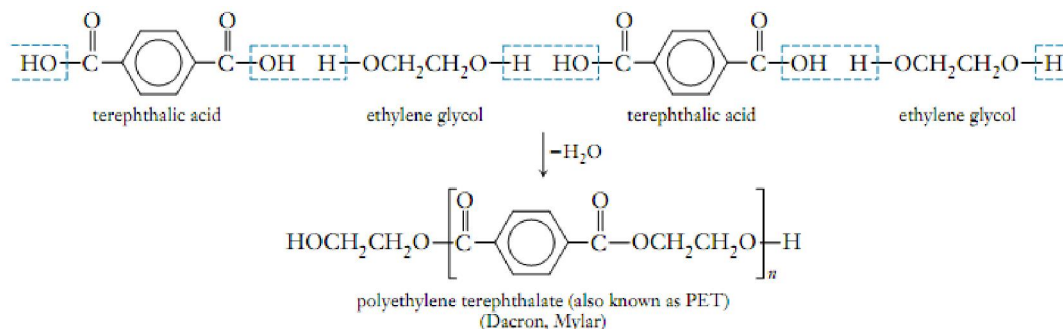
Some Important Addition Polymers

Polymer Name (some trade names)	Some Uses	Polymer Production, Tons/Year in United States	Monomer Name	Monomer Formula
polyethylene (Polythene)	electrical insulation; toys and molded objects; bags; squeeze bottles	14 million	ethylene	$\begin{array}{c} \text{H} & & \text{H} \\ & \backslash & / \\ & \text{C} = \text{C} \\ & / & \backslash \\ \text{H} & & \text{H} \end{array}$
polypropylene (Herculon, Vectra)	bottles; films; lab equipment, toys; packaging film; filament for rope, webbing, carpeting; molded auto and appliance parts	6.7 million	propylene	$\begin{array}{c} \text{H} & & \text{H} \\ & \backslash & / \\ & \text{C} = \text{C} \\ & / & \backslash \\ \text{H} & & \text{CH}_3 \end{array}$
polyvinyl chloride (PVC)	pipe, siding, gutters; floor tiles	7.4 million	vinyl chloride	$\begin{array}{c} \text{H} & & \text{H} \\ & \backslash & / \\ & \text{C} = \text{C} \\ & / & \backslash \\ \text{H} & & \text{Cl} \end{array}$
polyacrylonitrile (Orlon, Acrilan)	acrylic fibers for carpets, clothing, knitwear	95,000	acrylonitrile	$\begin{array}{c} \text{H} & & \text{H} \\ & \backslash & / \\ & \text{C} = \text{C} \\ & / & \backslash \\ \text{H} & & \text{CN} \end{array}$
polystyrene (Styrene, Styrofoam, Styron)	molded toys, dishes, kitchen equipment; insulating foam, e.g., ice chests; rigid foam packaging	3.3 million	styrene	$\begin{array}{c} \text{H} & & \text{H} \\ & \backslash & / \\ & \text{C} = \text{C} \\ & / & \backslash \\ \text{H} & & \text{C}_6\text{H}_5 \end{array}$
poly(methyl meth- acrylate) (Plexiglas, Lucite)	high-quality transparent objects; water- based paints; contact lenses	507,000	methyl methacrylate	$\begin{array}{c} \text{H} & & \text{CH}_3 \\ & \backslash & / \\ & \text{C} = \text{C} \\ & / & \backslash \\ \text{H} & & \text{C} = \text{O} \\ & & \\ & & \text{OCH}_3 \end{array}$
polybutadiene	automotive tire tread, hoses, belts; metal can coatings	610,000	1,3-butadiene	$\begin{array}{c} \text{H} & & \text{H} \\ & \backslash & / \\ & \text{C} = \text{C} \\ & / & \backslash \\ \text{H} & & \text{C} = \text{C} \\ & & / \quad \backslash \\ & & \text{H} \quad \text{H} \end{array}$
ethylene-propylene copolymer	appliance parts; auto hoses, bumpers, body and chassis parts; coated fabrics	324,000	ethylene, propylene	formulas given above
SBR copolymer	tires	1 million	styrene, 1,3-butadiene	formulas given above

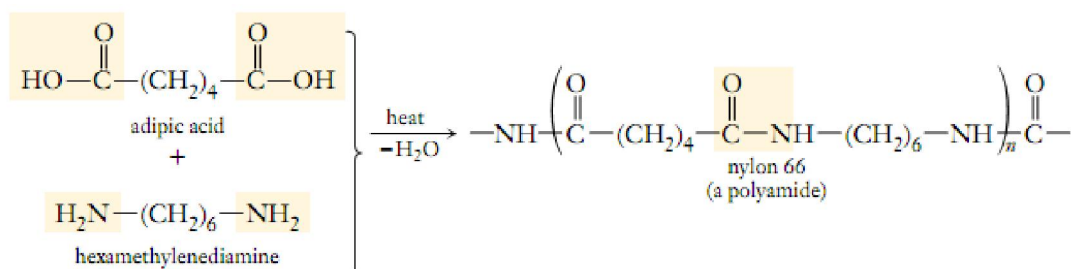
Condensation Polymerization

Some polymerization reactions are based on condensation reactions, in which two molecules combine by splitting out or eliminating a small molecule. For such a polymer to be formed, each monomer must have two functional groups, one on each end. A polymer formed in this way is called a condensation polymer. There are many useful condensation polymers, based on a wide variety of bifunctional molecules.

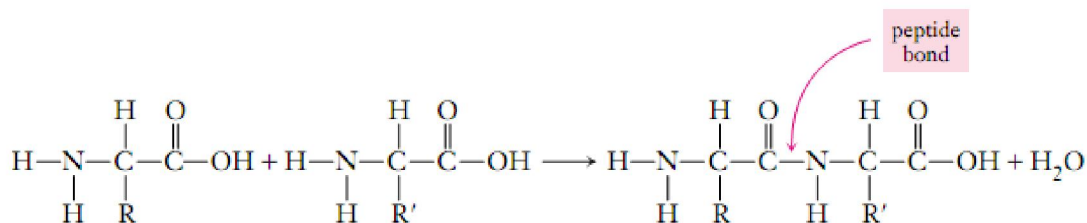
Polyesters (short for “polymeric esters”) are condensation polymers that are formed when dihydric alcohols react with dicarboxylic acids. An ester linkage is formed at each end of each monomer molecule to build up large molecules. A useful polyester is prepared from terephthalic acid and ethylene glycol.



Dacron, the fiber produced from this polyester, accounts for approximately 50% of all synthetic fibers. It absorbs very little moisture, and its properties are nearly the same whether it is wet or dry. Additionally, it possesses exceptional elastic recovery properties so it is used to make “permanent-press” fabrics. The polymeric amides, polyamides, are an especially important class of condensation polymers. Nylon is the best known polyamide. It is prepared by heating anhydrous hexamethylenediamine with anhydrous adipic acid, a dicarboxylic acid. This substance is often called nylon 66 because the parent diamine and dicarboxylic acid each contain six carbon atoms.



Some types of natural condensation polymers play crucial roles in living systems. Proteins are polymeric chains of L-amino acids linked by peptide bonds. A peptide bond is formed by the elimination of a molecule of water between the amino group of one amino acid and the carboxylic acid group of another.



When this process is carried out repeatedly, a large molecule called a polypeptide is formed.

